

Comparative efficacy and safety of PSCA CAR-engineered V δ 1 $\gamma\delta$ T cells for immunotherapy of pancreatic cancer

Zhixin Li,^{1,2} Ningyuan Liu,^{1,2} Zahir Shah,^{1,2} Lewei Jin,^{1,2} Lei Tian,^{1,2} Xiaolei Sun,^{1,2} Jianying Zhang,³ Zhiyao Li,^{1,2} Michael A Caligiuri ,^{1,2,4} Jianhua Yu^{5,6,7}

To cite: Li Z, Liu N, Shah Z, *et al.* Comparative efficacy and safety of PSCA CAR-engineered V δ 1 $\gamma\delta$ T cells for immunotherapy of pancreatic cancer. *Journal for ImmunoTherapy of Cancer* 2025;**13**:e011890. doi:10.1136/jitc-2025-011890

► Additional supplemental material is published online only. To view, please visit the journal online (<https://doi.org/10.1136/jitc-2025-011890>).

ZL, NL and ZS contributed equally.

Accepted 22 August 2025

ABSTRACT

Background $\gamma\delta$ T cells possess unique immunological features including tissue tropism, major histocompatibility complex-independent antigen recognition, and hybrid T/natural killer cell properties that make them promising candidates for cancer immunotherapy. However, the therapeutic potential of V δ 1 $\gamma\delta$ T cells, particularly when engineered with chimeric antigen receptors (CARs), remains underexplored in solid tumors such as pancreatic cancer (PC), largely due to their low abundance in peripheral blood and challenges in ex vivo expansion. This study aims to directly compare the preclinical safety and efficacy among CAR-engineered V δ 1 $\gamma\delta$ T cells, V δ 2 $\gamma\delta$ T cells, and conventional $\alpha\beta$ T cells.

Methods We developed a CAR targeting prostate stem cell antigen (PSCA) and engineered it into human blood-derived V δ 1 $\gamma\delta$ T cells. These PSCA CAR-V δ 1 T cells were evaluated for antitumor activity against PSCA-expressing pancreatic tumor cells using both in vitro cytotoxicity assays and in vivo xenograft mouse models. Comparative analyses were conducted using PSCA CAR-V δ 1 $\gamma\delta$ T cells, PSCA CAR-V δ 2 $\gamma\delta$ T cells, and PSCA CAR- $\alpha\beta$ T cells. Safety assessments of these CAR-engineered T cell subsets were conducted to evaluate graft-versus-host disease (GvHD) and cytokine release syndrome (CRS), while exhaustion profiles were assessed using single-cell RNA sequencing.

Results Our study demonstrated that PSCA CAR-V δ 1 $\gamma\delta$ T cells, PSCA CAR-V δ 2 $\gamma\delta$ T cells, and PSCA CAR- $\alpha\beta$ T cells exhibited similar antitumor efficacy. However, PSCA CAR-V δ 1 T cells did not exhibit GvHD or CRS compared with CAR- $\alpha\beta$ T cells. PSCA CAR-V δ 1 T cells also had lower scores for exhaustion when compared with PSCA CAR-V δ 2 T cells by single-cell transcriptomic analysis.

Conclusion PSCA CAR-V δ 1 $\gamma\delta$ T cells represent a potent and safe therapeutic modality for PSCA⁺ PC, with efficacy comparable to other CAR T cell types and potential advantages in safety and persistence. These findings support further development of CAR-V δ 1 T cell immunotherapy for solid tumors.

INTRODUCTION

Pancreatic cancer (PC) currently ranks as the fourth leading cause of cancer-related mortality in the Western world, with a 5-year survival rate of approximately 10%.¹ By 2030, PC is projected to surpass breast, prostate,

WHAT IS ALREADY KNOWN ON THIS TOPIC

- ⇒ $\gamma\delta$ T cells have distinctive immune features including tissue tropism, major histocompatibility complex-independent antigen recognition, and hybrid T/natural killer cell functionality.
- ⇒ Compared to conventional $\alpha\beta$ T cells, $\gamma\delta$ T cells possess a safer profile with regards to graft-versus-host disease (GvHD) and cytokine release syndrome (CRS).

WHAT THIS STUDY ADDS

- ⇒ This study directly compares the efficacy of prostate stem cell antigen (PSCA) chimeric antigen receptor (CAR)-V δ 1 $\gamma\delta$ T cells with PSCA CAR-V δ 2 $\gamma\delta$ T cells and PSCA CAR- $\alpha\beta$ T cells in both in vitro and in vivo pancreatic cancer models.
- ⇒ PSCA CAR-V δ 1 $\gamma\delta$ T cells demonstrate antitumor efficacy comparable to PSCA CAR-V δ 2 $\gamma\delta$ T cells and PSCA CAR- $\alpha\beta$ T cells.
- ⇒ Unlike PSCA CAR- $\alpha\beta$ T cells, PSCA CAR-V δ 1 $\gamma\delta$ T cells do not induce GvHD or CRS and exhibit a lower exhaustion profile when compared to CAR-V δ 2 $\gamma\delta$ T cells.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

- ⇒ This study highlights PSCA CAR-V δ 1 T cells as an equally efficacious but safer alternative to PSCA CAR- $\alpha\beta$ T cells for the treatment of PSCA⁺ solid tumors, with demonstrated capability for allogeneic use.

and colorectal cancers to become the second leading cause of cancer-related deaths after lung cancer.² Nearly 90% of pancreatic malignancies are classified as pancreatic ductal adenocarcinomas, which are typically diagnosed at an advanced, incurable stage due to the absence of effective screening methods and the scarcity of specific early symptoms. Since 1997, gemcitabine has been an option for treatment in locally advanced and metastatic PC.³ However, it has only a modest effect on survival, as gemcitabine chemoresistance can develop within a few weeks of treatment in initially chemosensitive tumors.⁴ Moreover,



© Author(s) (or their employer(s)) 2025. Re-use permitted under CC BY-NC. No commercial re-use. See rights and permissions. Published by BMJ Group.

For numbered affiliations see end of article.

Correspondence to
Dr Michael A Caligiuri;
mcaligiuri@coh.org

Jianhua Yu; jianhuay@uci.edu

among other treatment modalities, PC has been largely unresponsive to immune checkpoint blockade therapy, unless it carries the rare alterations in mismatch repair.⁵ Effective treatment options are thus still limited for PC, and as such, there is an urgent need for innovative and safe treatment modalities for advanced PC.

$\gamma\delta$ T cells constitute a minor subset of T lymphocytes that exhibit characteristics of both adaptive and innate immunity. Their subsets are distinguished by the δ chain of their T cell receptor (TCR), with V δ 1-expressing $\gamma\delta$ T cells predominantly residing in mucosal tissues, while V δ 2-expressing cells are mainly found in the peripheral blood.^{6,7} Unlike conventional $\alpha\beta$ T cells, which recognize peptide antigens presented by polymorphic major histocompatibility complex (MHC) class I and II molecules, $\gamma\delta$ T cells recognize invariant antigens in an MHC-independent manner. This unique recognition mechanism positions $\gamma\delta$ T cells as promising candidates for the development of allogeneic, off-the-shelf cell-based immunotherapies for cancer. Furthermore, tumor-infiltrating $\gamma\delta$ T cells are positively associated with patient survival across diverse cancers.^{8,9} Early clinical investigations of $\gamma\delta$ T cells have shown antitumor effects on hematologic cancers as well as solid tumors with limited adverse effects in vivo.^{10–12} In 2007, Kobayashi *et al.* reported the first pilot clinical study of in vitro expanded autologous $\gamma\delta$ T cell therapy for advanced renal cell carcinoma; however, all patients (7 out of 7) eventually showed progressive disease.¹³ Other early clinical studies experienced a high rate of progressive disease and inconsistent results, demonstrating a need to increase $\gamma\delta$ T cell antitumor activity.^{14–17} Given the success of chimeric antigen receptor (CAR)-T cell therapy using conventional $\alpha\beta$ T cell therapy in the clinic for hematologic malignancies,^{18,19} investigators have co-opted this strategy to engineer and test CAR $\gamma\delta$ T cell therapies in both liquid and solid tumors.^{20–24} While a few $\gamma\delta$ T cell-related immunotherapies without a CAR have been reported as effective for PC treatment,^{25,26} the use and therapeutic effects of CAR $\gamma\delta$ T cells especially V δ 1 $\gamma\delta$ T cells (hereafter referred to as V δ 1 T cells) for PC have not been investigated.

The clinical application of $\gamma\delta$ T cells has been somewhat limited to the predominant circulating V δ 2 $\gamma\delta$ T subset (hereafter referred to as V δ 2 T cells). This is in part due to the discovery that soluble phosphoantigens (eg, isopentenyl pyrophosphate (IPP), (E)-4-hydroxy-3-methyl-but-2-enyl pyrophosphate (HMBPP), or bromohydrin pyrophosphate (BrHPP))^{27–29} or phosphoantigen-inducing aminobisphosphonates (eg, zoledronic acid) can be used to expand V δ 2 T cells to high numbers with excellent purity in vitro,^{30,31} which accelerated the clinical translation of V δ 2 T cells. Although no V δ 1 TCR cognate ligand has been reported for V δ 1 T cell specific expansion, two clinical-grade protocols for V δ 1 cell in vitro expansion reported by Almeida³² and Ferry³³ may provide the opportunity to accelerate clinical development of V δ 1 T cell-based immunotherapy. Studies have shown that V δ 1 T cells are less prone to exhaustion

compared with their V δ 2 T counterparts, which might provide a therapy with more persistence.^{34,35} Furthermore, even within the milieu of exhaustion, intratumoral V δ 2[–] $\gamma\delta$ T cells (ie, mainly V δ 1 T cells) in human kidney cancer that exhibit markers of terminal T cell exhaustion have been found to still maintain cytotoxic function.^{36,37} This suggests that such cells may provide a more effective cancer immunotherapeutic option compared with their V δ 2 T counterparts.

Prostate stem cell antigen (PSCA) is a glycosylphosphatidylinositol-anchored cell surface protein that plays a role in intracellular signaling and the proliferation of prostate tumors.³⁸ Aberrant overexpression of PSCA has been observed in 60–80% of PC tumors,^{39,40} while its expression remains limited in normal tissues.⁴¹ Thus, PSCA is a potential target for PSCA CAR immunotherapy, and its safety and efficacy have been evaluated recently with CAR-T cells, CAR-natural killer (NK) cells, and CAR-macrophages.^{42–44} To explore the tumor reactivity and safety profile of V δ 1 T cells, we developed human PSCA CAR-V δ 1 T cells harboring a truncated epidermal growth factor receptor (tEGFR) as a safety switch⁴³ and evaluated the product in vitro and in vivo. Furthermore, we have conducted a parallel comparison of V δ 1, V δ 2, and conventional $\alpha\beta$ T cells engineered with PSCA CAR against PC. Our PSCA CAR-V δ 1 T cells show potent CAR-specific antitumor activity against PC both in vitro and in vivo, leading to prominent tumor regression and prolonged survival without obvious graft-versus-host disease (GvHD) or systemic toxicity. Our preclinical results suggest that PSCA CAR-V δ 1 T cell-based cellular therapy may present an attractive option for clinical development against PC.

RESULTS

Generation and immunophenotype of peripheral blood-derived V δ 1 T cells

V δ 1 and V δ 2 T cells and $\alpha\beta$ T cells were isolated, activated and expanded from healthy donors' peripheral blood mononuclear cells (PBMCs) as described in Methods (figure 1A). Flow cytometric analysis demonstrated that during 14 days of in vitro culture, both V δ 1 and V δ 2 T cells were successfully enriched (online supplemental figure 1A). Our V δ 1 T cell enrichment and expansion protocol resulted in a significant increase in V δ 1 T cells, from 0.4% to 9.1% of PBMCs after 21 days of enrichment and expansion (online supplemental figure 1B), representing a more than 1500-fold increase in the absolute number of V δ 1 T cells (figure 1B and C). Compared with pre-expanded V δ 2 and $\alpha\beta$ T cells, pre-expanded V δ 1 T cells exhibited lower activation and effector memory profiles but a higher proportion of effector T cell phenotypes (figure 1D). On expansion, V δ 1 T cells demonstrated markedly elevated levels of NK cell-activating receptors such as NKG2D⁴⁵ and CD25,⁴⁵ the checkpoint receptor programmed cell death protein-1 (PD-1) with effector response,³⁷ and a pronounced effector memory

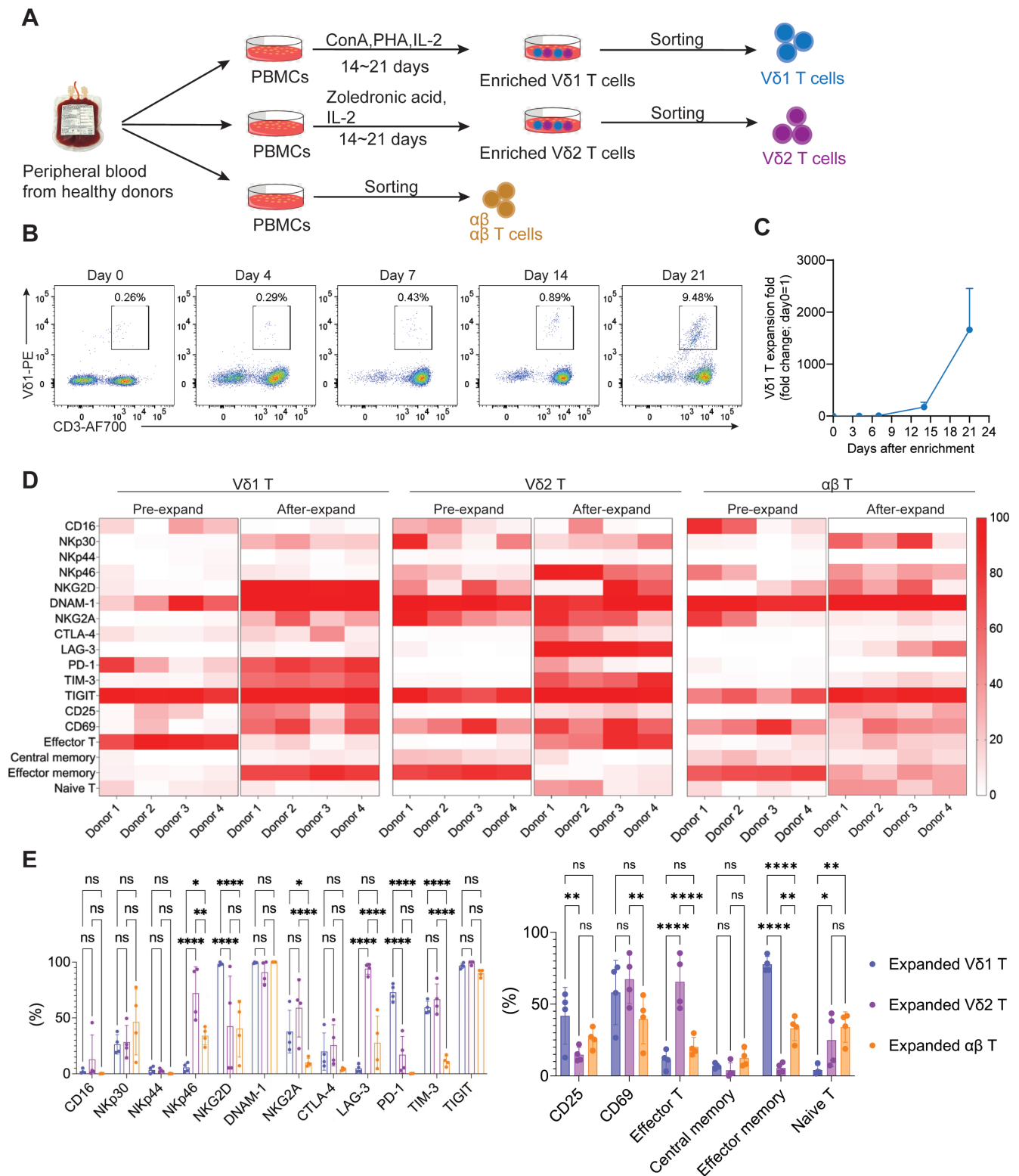


Figure 1 Generation and characterization of T cell subsets from peripheral blood. (A) Conceptual diagram depicting in vitro enrichment and expansion protocol for Vδ1 T cells, Vδ2 T cells, and αβ T cells isolated from PBMCs. (B) Representative flow cytometry dot plots of Vδ1 enrichment at different time points. (C) Expansion fold change of Vδ1 T cells over 3 weeks from (B) (n=5). (D) Phenotypic comparison of Vδ1, Vδ2 T, and αβT cells before and after expansion of each from the same donors. Data were collected by flow cytometry and presented as heatmaps for comparison (single gradient colormap, largest value (100) in bright red, lowest value (0) in white, Scale interval 20) (n=4 normal donors). Effector T cell phenotype is identified by CD45RO⁻CCR7⁻, central memory T cells by CD45RO⁺CCR7⁺, effector memory T cells by CD45RO⁺CCR7⁻, and naive T cells by CD45RO⁻CCR7⁺. (E) Summary data presented in (D). ns, not significant or p>0.05, *p<0.05, **p<0.01, ***p<0.001 (one-way ANOVA). ANOVA, analysis of variance; ConA, Concanavalin A; IL-2, interleukin 2; PBMCs, peripheral blood mononuclear cells; PD-1, programmed cell death protein-1; PHA, phytohemagglutinin.

phenotype surpassing those of expanded V δ 2 and $\alpha\beta$ T cells (figure 1D and E, online supplemental figure 1C). These findings indicate a robust activation and memory profile of V δ 1 T cells following in vitro expansion.

Next, we compared the phenotype of expanded V δ 1 T cells to expanded V δ 2 T cells and expanded conventional $\alpha\beta$ T cells. Although expanded V δ 2 T cells were also activated, they expressed relatively higher levels of inhibitory receptors LAG-3 and NKG2A, suggesting a more suppressive or exhausted phenotype. $\alpha\beta$ T cells, by comparison, generally exhibited lower expression of NK cell-associated markers and distinct memory subset distributions, highlighting the functional plasticity of $\gamma\delta$ T cells (figure 1D and E). The surface density of additional markers including CD16, NKp44, DNAM-1, NKG2A, CTLA-4, TIM-3, TIGIT, and CD69 was broadly comparable between V δ 1 and V δ 2 T cells, though subject to interdonor variability (figure 1E).

Collectively, these results support a more balanced immunophenotype for V δ 1 T cells, characterized by strong expression of antigens signifying cytotoxic potential (eg, NKG2D), antigens regulating immune checkpoint (eg, PD-1), and retention of memory subsets; features consistent with a versatile, tumor-reactive, and potentially checkpoint-responsive profile. Notably, expression patterns of such antigens remain consistent across donors, underscoring a reproducible trend despite inter-individual variation.

PSCA CAR-V δ 1 T cells exhibit CAR-specific serial lysis of pancreatic cancer cells in vitro

To generate CAR-V δ 1 T cells, enriched V δ 1 T cells ($\geq 95\%$ purity) were transduced with a retrovirus-based PSCA CAR construct (figure 2A), previously validated for its PSCA-specific antitumor efficacy in CAR-NK cells and CAR-macrophages.^{43–46} To assess the PSCA CAR-dependent cytotoxicity against pancreatic cancer (PC) cells in vitro, we selected PSCA-high expressing (PSCA^{high}) tumor cells (Capan-1 and MIA PaCa-2) and PSCA-low expressing (PSCA^{low}) tumor cells (PANC-1) for functional validation (figure 2B). Prior to co-culture with PC cells, the purity of PSCA CAR-V δ 1 T cell exceeded 95% and CAR expression, as determined by expression of the tEGFR marker, ranged from 40–60% across all donors (figure 2C and D, online supplemental figure 2A). Subsequently, both PSCA^{high} and PSCA^{low} PC cells were cocultured for 24 hours with PSCA CAR-V δ 1 T cells or mock-V δ 1 T cells (mock consisted of the transduced tEGFR without the PSCA CAR) (figure 2E). The PSCA CAR-V δ 1 T cells exhibited significantly higher cytotoxicity compared with mock-transduced V δ 1 T cells against PSCA^{high} Capan-1 and PSCA^{high} MIA PaCa-2 tumor cells, but not against PSCA^{low} PANC-1 tumor cells (figure 2F), demonstrating PSCA CAR-dependent killing of PSCA-positive tumor cells. Of note, both fresh and cryopreserved-thawed PSCA CAR-V δ 1 T cells demonstrated comparable cytolytic potency against Capan-1 cells, highlighting their potential as an

effective off-the-shelf therapeutic option (online supplemental figure 2B and C).

Having established the CAR-specific antitumor functionality of activated PSCA CAR-V δ 1 T cells, we proceeded to assess their capability for serial killing of tumor cells using the real-time cytotoxicity assay (RTCA).^{47–49} Activated PSCA CAR-V δ 1 T cells underwent repeated exposure to target cells in a serial killing assay with the initial effector/target (E:T) ratio of 1:2, where Capan-1 target cells, but not effector cells, were replenished on two additional occasions within 7 days (online supplemental figure 2D) while the tumor cell growth index was measured as previously described.⁵⁰ Results measured over time provide evidence of serial tumor killing by PSCA CAR-V δ 1 T cells in vitro (online supplemental figure 2D). The enhanced cytolytic function of PSCA CAR-V δ 1 T cells was further confirmed by their elevated release of granzyme B (GZMB) and interferon-gamma (IFN- γ) that were significantly higher when cocultured with PSCA^{high} tumor cells, compared with mock transduced V δ 1 T cells (online supplemental figure 2E). To further explore the key mediators of this cytotoxic property, we blocked NKG2D, DNAM-1, or NKp30 on PSCA CAR-V δ 1 T cells but did not observe a decrease in cytotoxicity of PSCA^{high} tumor cells, underlining the dominant role of the PSCA CAR in the antitumor activity of PSCA CAR-V δ 1 T cells (online supplemental figure 2F). In summary, the CAR engineering results in enhanced cytotoxicity of PSCA CAR-V δ 1 T cells against PSCA-expressing PC tumor cells, perhaps mediated at least in part by GZMB and IFN- γ secretion (online supplemental figure 2G). PSCA CAR-V δ 1 T cells also demonstrate promising serial killing capacity against PSCA^{high} Capan-1 cells. Comparable antitumor efficacy between fresh and cryopreserved PSCA CAR-V δ 1 T cells highlighted the PSCA CAR-V δ 1 T cells as a promising candidate for off-the-shelf immunotherapy.

Next, we compared functional cytotoxicity of expanded un-engineered and PSCA CAR-engineered V δ 1, V δ 2, and conventional $\alpha\beta$ T cells in vitro (figure 2 and online supplemental figure 3). Non-transduced and CAR-transduced V δ 1, V δ 2, and $\alpha\beta$ T cells were co-cultured with Capan-1 cells (online supplemental figure 3E). Cytotoxicity assays revealed no significant difference in capacity for tumor-killing among the three T cell subsets, with or without CAR expression (figure 2G and H, online supplemental figure 3F and G). We also saw no significant differences in the IFN- γ expression among these groups. However, CAR-V δ 1 and CAR-V δ 2 T cells exhibited notably higher levels of GZMB compared with CAR- $\alpha\beta$ T cells (figure 2I). These data indicate that V δ 1 T cells exhibit antigen-specific cytotoxic activity at least comparable to that of V δ 2 and $\alpha\beta$ T cells.

PSCA CAR-V δ 1 T cells display persistent tumor control and tumor infiltration in vivo

To evaluate the therapeutic efficacy of PSCA CAR-V δ 1 T cells in vivo, we established a metastatic PC xenograft model in NSG mice. Cryopreserved PSCA CAR-V δ 1 T

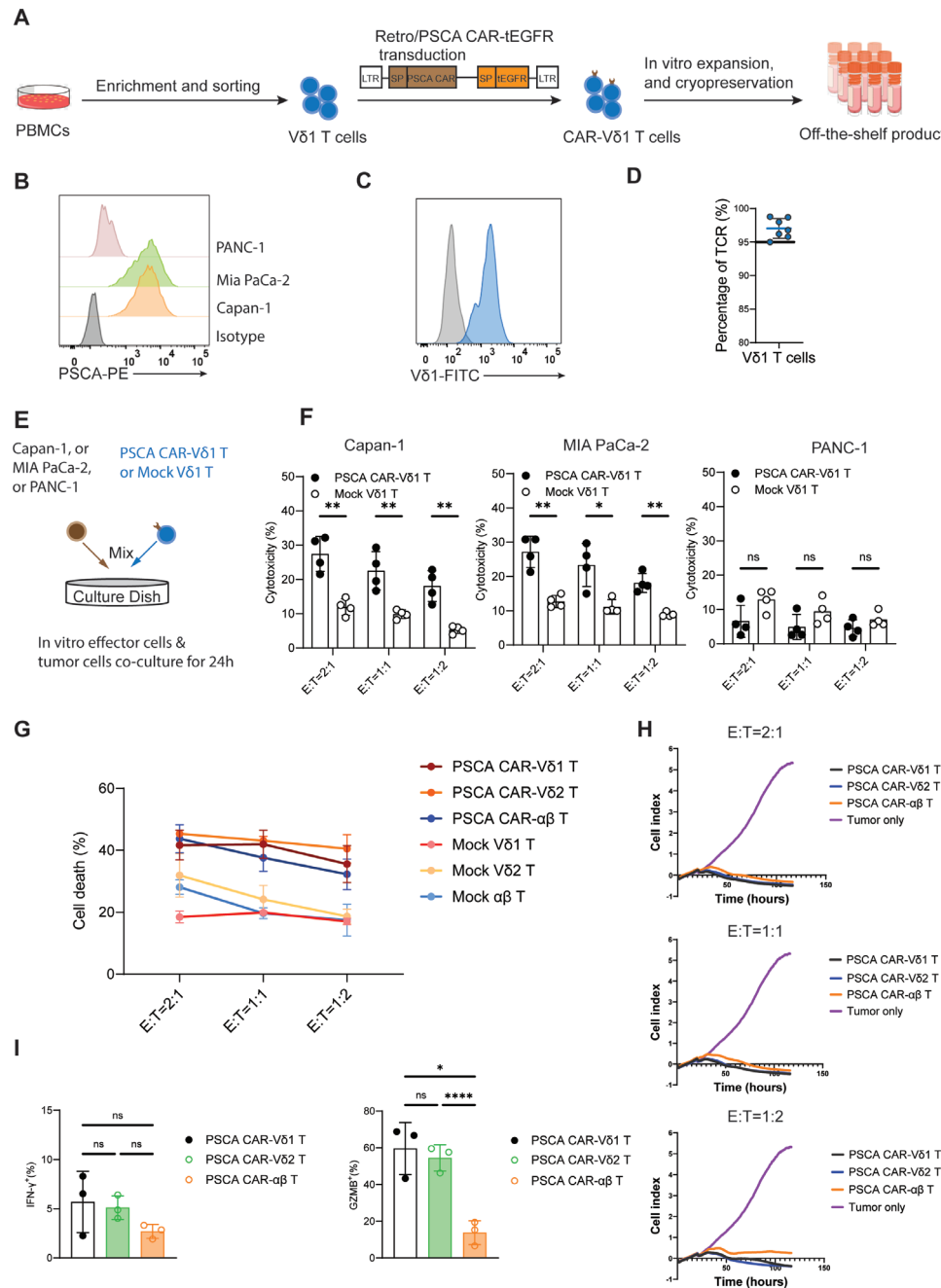


Figure 2 Assessment of functional responses of PSCA CAR-Vδ1 T, PSCA CAR-Vδ2 T, and PSCA CAR-αβ T cells to PSCA^{high} tumor cells in vitro. (A) Schematic representation of the process for manufacturing off-the-shelf allogeneic PSCA CAR-Vδ1 T cells. (B) Median fluorescence intensity (MFI) for PSCA surface expressions on the indicated PC cell lines. (C) MFI of Vδ1⁺ γδ T cells following expansion. (D) Vδ1⁺ γδ T cell purity (n=7). (E) Schema of PSCA CAR-Vδ1 T cells or mock-transduced Vδ1 T cells cocultured with different PC cells at an effector to target (E:T) ratio of 2:1 (10,000 cells to 5,000 cells), 1:1 (5,000 cells to 5,000 cells), and 1:2 (2,500 cells to 5,000 cells) for 24 hours (n=4, triplicate wells). (F) Cytotoxicity of PSCA CAR-Vδ1 T cells against PSCA^{high} Capan-1 cells and PSCA^{high} MIA PaCa-2 cells and PSCA^{low} PANC-1 cells at E:T ratio of 2:1, 1:1, and 1:2 measured after 24 hours co-culture (n=4); ns, not significant. p>0.05, *p<0.05, **p<0.01 (unpaired t test). (G) Representative data showing the percentage of Capan-1 cell death in an Annexin V/DAPI assay after 24-hour co-culture at the indicated effector-to-target (E:T) ratios with mock-engineered or PSCA CAR-engineered Vδ1, Vδ2, or αβ T effector cells (n=3). (H) Killing assay was performed using a real-time cytotoxicity assay with an E:T ratio of 1:1 (4,000 PSCA CAR-Vδ1 T or PSCA CAR-Vδ2 T or PSCA CAR-αβ T effector cells to 4000 Capan-1 cells added at each of the three indicated intervals) and presented as the growth index of the residual Capan-1 cells (n=3). (I) Representative and summary data of IFN-γ (left) and granzyme B (GZMB; right) expression in PSCA CAR-Vδ1 T cells compared with PSCA CAR-Vδ2 T cells and PSCA CAR-αβ T cells, after being cocultured with PSCA^{high} Capan-1 cells with E:T ratio of 1:1 (n=3); *p<0.05, ****p<0.0001 (t test). CAR, chimeric antigen receptor; DAPI, 4',6-diamidino-2-phenylindole; IFN-γ, interferon-gamma; LTR, Long Terminal Repeat; PBMCs, peripheral blood mononuclear cells; PC, pancreatic cancer; PSCA, prostate stem cell antigen; SP, Spacer Peptide; TCR, T cell receptor; tEGFR, truncated estimated glomerular filtration rate.

cells were thawed and infused as shown in [figure 3A](#). PSCA CAR-V δ 1 T cells showed superior therapeutic benefit compared with the saline and mock-cryopreserved V δ 1 T cell groups by inhibiting the progression of metastatic PC in a dose-dependent manner as determined by monitoring bioluminescence imaging (BLI) in live animals over time and quantification of the BLI ([figure 3B and C](#)). Immunohistochemical staining revealed negligible PSCA protein expression in the liver, lung, pancreas, and spleen following treatment with PSCA CAR-V δ 1 T cells, suggesting effective tumor cell eradication (online supplemental figure 3A). To confirm that the absence of PSCA after PSCA CAR-V δ 1 T cell treatment represented tumor eradication rather than antigen loss, we stained the same samples with MUC1, a PC marker, and it showed a similar staining profile like that of PSCA (online supplemental figure 3B). Hematoxylin and eosin (H&E) staining also demonstrated no obvious damage to normal tissues after PSCA CAR-V δ 1 T cell treatment (online supplemental figure 3C). Furthermore, administration of a high dose (6×10^6 PSCA CAR-V δ 1 T cells per infusion, delivered every other week for a total of three infusions, with each dose divided between intraperitoneal (i.p.) and intravenous routes for experimental and control groups) significantly extended the survival of treated mice compared with those receiving a low dose of PSCA CAR-V δ 1 T cells (3×10^6 PSCA CAR-V δ 1 T cells per infusion) or mock-transduced V δ 1 T cells, or saline ([figure 3D](#)).

PC is characterized by dense fibrotic stroma, or desmoplasia, which presents a major barrier to effective infiltration of adoptive cell therapies into solid tumors. To further understand the mechanisms underlying the effectiveness of PSCA CAR-V δ 1 T cell therapy, including tissue-specific effects, we established an orthotopic mouse model to mimic the natural microenvironment and tumor growth characteristics of cancer within the pancreas. We implanted the PC cell line Capan-1_Luc_ZsGreen into the pancreas of NSG mice and intravenously infused a relatively low dose of PSCA CAR-V δ 1 T cells to treat the cancer and used mock transduced V δ 1 T cells as a control ([figure 3E](#)). Ten days after V δ 1 T cell infusion, mice were humanely sacrificed for evidence of infiltration into normal pancreatic tissues and PC tissues ([figure 3E](#)). Our analysis demonstrated that intravenously injected PSCA CAR-V δ 1 T cells can effectively infiltrate into the pancreas and tumor, as evidenced by flow cytometric analysis, while mock transduced V δ 1 T cells did not ([figure 3F and G](#)), again highlighting the importance of the PSCA CAR V δ 1 T cells.

In vivo benchmarking of V δ 1, V δ 2, and conventional $\alpha\beta$ T cells with PSCA CAR reveals comparable tumor control in pancreatic cancer

Despite several clinical trials exploring V δ 1 and/or V δ 2 cells and their CAR-engineered derivatives for cancer therapy (NCT05001451, NCT04735471, NCT03533816, and NCT04107142), no preclinical head-to-head comparison of safety and antitumor efficacy among V δ 1, V δ 2, and

conventional $\alpha\beta$ T cells has been conducted. To address this gap, we performed a preclinical comparison of V δ 1, V δ 2, and $\alpha\beta$ T cells with and without PSCA CAR transduction using two distinct solid tumor models ([figure 4A and E](#)). We first assessed the antitumor efficacy of non-transduced V δ 1, V δ 2, and $\alpha\beta$ T cells in a subcutaneous (s.c.) NSG mouse model implanted with Capan-1_Luc_ZsGreen PC cells ([figure 4A](#)). Following two intravenous infusions of each T cell type, V δ 1 T cell-treated mice exhibited a favorable, though not statistically significant, trend toward tumor growth suppression relative to saline-, V δ 2-, and $\alpha\beta$ T cell-treated groups ([figure 4B–D](#)).

Subsequently, we evaluated the *in vivo* antitumor activity of PSCA CAR-V δ 1, PSCA CAR-V δ 2, and PSCA CAR- $\alpha\beta$ T cells in a metastatic PC model established via intraperitoneal injection of Capan-1 cells ([figure 4E](#)). A single administration of these CAR-engineered T cells (via both intravenous and i.p. routes) significantly controlled tumor growth and delayed disease progression compared with saline controls ([figure 4F–H](#)), resulting in durable remission and, in some cases, complete regression with no discernible differences in efficacy among the three CAR-T cell types ([figure 4F1](#)). These findings were corroborated by our *in vitro* cytotoxicity assays ([figure 2G and H](#); Online supplemental Figure 3F and G).

Overall, PSCA CAR-V δ 1 T cells demonstrated robust and comparable cytotoxic activity both *in vitro* and *in vivo*, relative to CAR-V δ 2 and CAR- $\alpha\beta$ T cells, within our PC models.

Single-cell transcriptional profiling of PSCA CAR-V δ 1 and V δ 2 T cells reveals distinct profiles

We next performed single-cell RNA sequencing (scRNA-seq) to profile PSCA CAR-V δ 1, PSCA CAR-V δ 2, and PSCA CAR- $\alpha\beta$ T cells following adoptive transfer (hereafter referred to as *in vivo*). An orthotopic model of PC was established in NSG mice by implanting Capan-1 PC cells into the pancreas. Seven days after intravenous and i.p. injection of PSCA CAR-V δ 1, PSCA CAR-V δ 2, or PSCA CAR- $\alpha\beta$ T cells, each T cell subset was harvested for scRNA-seq analysis. The corresponding preinfusion PSCA CAR-V δ 1, PSCA CAR-V δ 2, and PSCA CAR- $\alpha\beta$ T cell products served as *in vitro* controls. To minimize intersample variability and enable unbiased group comparisons, each sample was pooled from two independent tumor-bearing mice ([figure 5A](#)). Data visualization using uniform manifold approximation and projection (UMAP),⁵¹ combined with unsupervised clustering, identified three principal cell types: PSCA CAR-V δ 1, PSCA CAR-V δ 2, and PSCA CAR- $\alpha\beta$ T cells ([figure 5B](#)). Different T cell subsets (V δ 1, V δ 2, and $\alpha\beta$ T cells) were recognized based on their TCR gene expression ([figure 5C](#)). Functional cluster analysis of PSCA CAR-V δ 1, PSCA CAR-V δ 2, and PSCA CAR- $\alpha\beta$ T cells revealed four previously characterized cellular states: (1) cytolytic/effector, (2) exhausted, (3) naïve, and (4) proliferative, each differentially varied among these CAR-engineered T cell subsets ([figure 5D and E](#)).⁵² Further analysis of the PSCA CAR-V δ 1 T cells found that

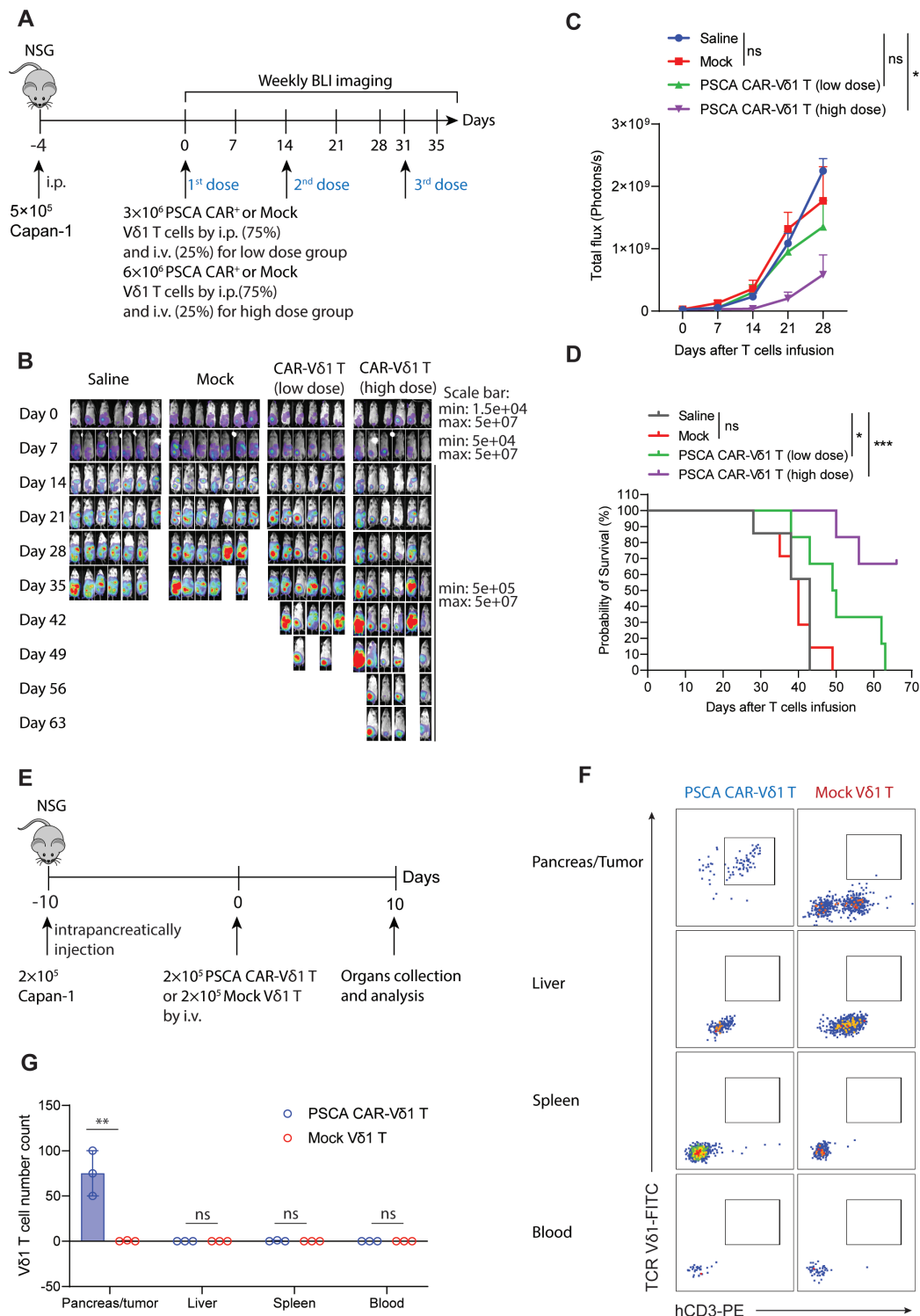


Figure 3 PSCA CAR-Vδ1 T cells suppress PC growth in vivo and exhibit pancreas/tumor infiltration. (A) A schematic of the treatment schedule for PSCA CAR-Vδ1 T cells or mock-transduced Vδ1 T cells in a human metastatic PC model generated by intraperitoneally injecting 0.5×10^6 Capan-1_Luc_ZsGreen cells into each NSG mouse. (B) Time-lapse luciferase imaging of the metastatic PC mouse model after the indicated treatments, $n=6-7$. Effector cells were infused by intravenous (25% of dose) and i.p. (75% of dose). (C) Quantification of the bioluminescence images from (B) up to day 28 ($n=4-6$). * $p < 0.01$ (one-way ANOVA). (D) Survival analysis of mice in the indicated groups from (B). $n=6-7$; * $p < 0.05$, *** $p < 0.001$ (log-rank (Mantel-Cox) test). (E) A schematic of treatment schedule for PSCA CAR-Vδ1 T cells in an orthotopic mouse model of human PC. (F) FACS analysis of the PSCA CAR-Vδ1 T cells or mock transduced Vδ1 T cell infiltration in the orthotopic mouse model of human PC. Pancreas/tumor, liver, spleen, and peripheral blood were collected at the end of the trial and subjected to FACS analysis. (G) Summary of (F). $n=3$. ns, not significant, ** $p < 0.01$ (t test). ANOVA, analysis of variance; BLI, bioluminescence imaging; CAR, chimeric antigen receptor; FACS, Fluorescence-activated cell sorting; i.p., intraperitoneal; i.v., intravenous; PC, pancreatic cancer; PSCA, prostate stem cell antigen.

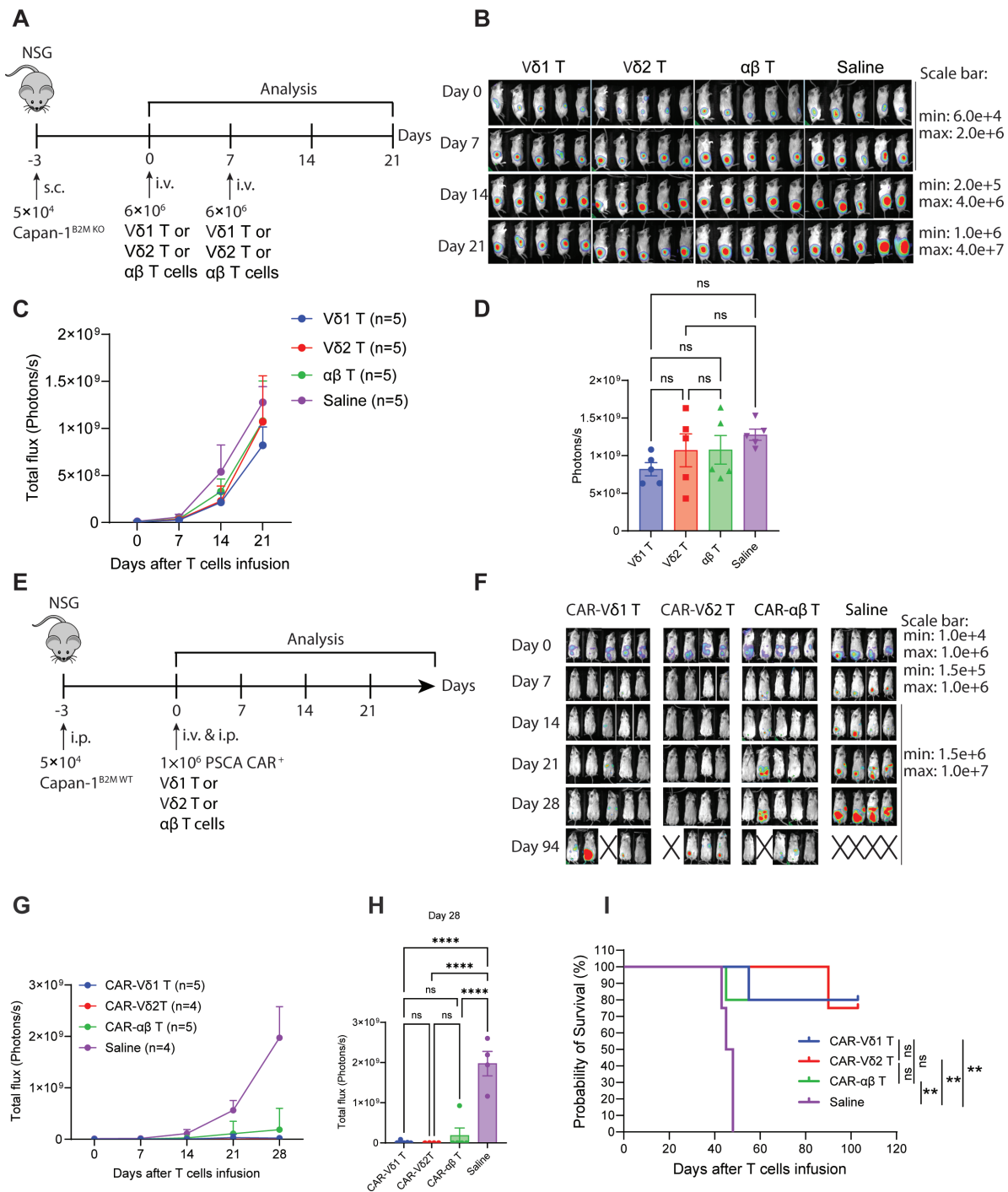


Figure 4 Head-to-head in vivo comparison of antitumor efficacy between Vδ1, Vδ2 and αβ T cells without and with PSCA CAR transduction. (A) Experimental design for in vivo comparison of tumor suppression between different subsets of T cells without PSCA CAR engineering. The tumor model was established by subcutaneous (s.c.) injection of 5×10^4 Capan-1^{Luc_sGreen} cells. Vδ1 or Vδ2 or αβ T cells were FACS sorted from 2 to 3 independent donors for each group. (B) BLI images showing tumor burden in experimental mice over time as shown in (A). (C) Quantification of BLI from (B) (n=5). (D) Tumor size measurements on day 21 from (B) (n=5), mean values of each group are plotted on the top of each column, ns, not significant. (E) Experimental design for in vivo comparison of tumor suppression between Vδ1 and Vδ2 subsets of γδ T cells and αβ T cells transduced with PSCA CAR. Effector cells were infused by intravenous (25% of dose) and i.p. (75% of dose). PSCA CAR-Vδ1 or PSCA CAR-Vδ2 or PSCA CAR-αβ T cells were FACS sorted to enrich for 100% CAR-expressing cells before infusion into mice. Effector cells from 2 to 3 independent donors were used for each group. (F) BLI images showing tumor loads in experimental mice over time. (G) Quantification of (F) (n=5 for CAR-Vδ1 T cell group and CAR-αβ T cell group, n=4 for CAR-Vδ2 T cell group and saline group) ****p<0.0001 (one-way ANOVA). (H) Summary of tumor size measurements in photons on day 28 from (G) (n=4-5), ****p<0.0001 (one-way ANOVA). (I) Survival analysis of mice in the indicated groups from (F). n=4-5. **p<0.01 (log-rank (Mantel-Cox) test). ANOVA, analysis of variance; BLI, bioluminescence imaging; CAR, chimeric antigen receptor; FACS, Fluorescence-activated cell sorting; i.p., intraperitoneal; i.v., intravenous; PSCA, prostate stem cell antigen.

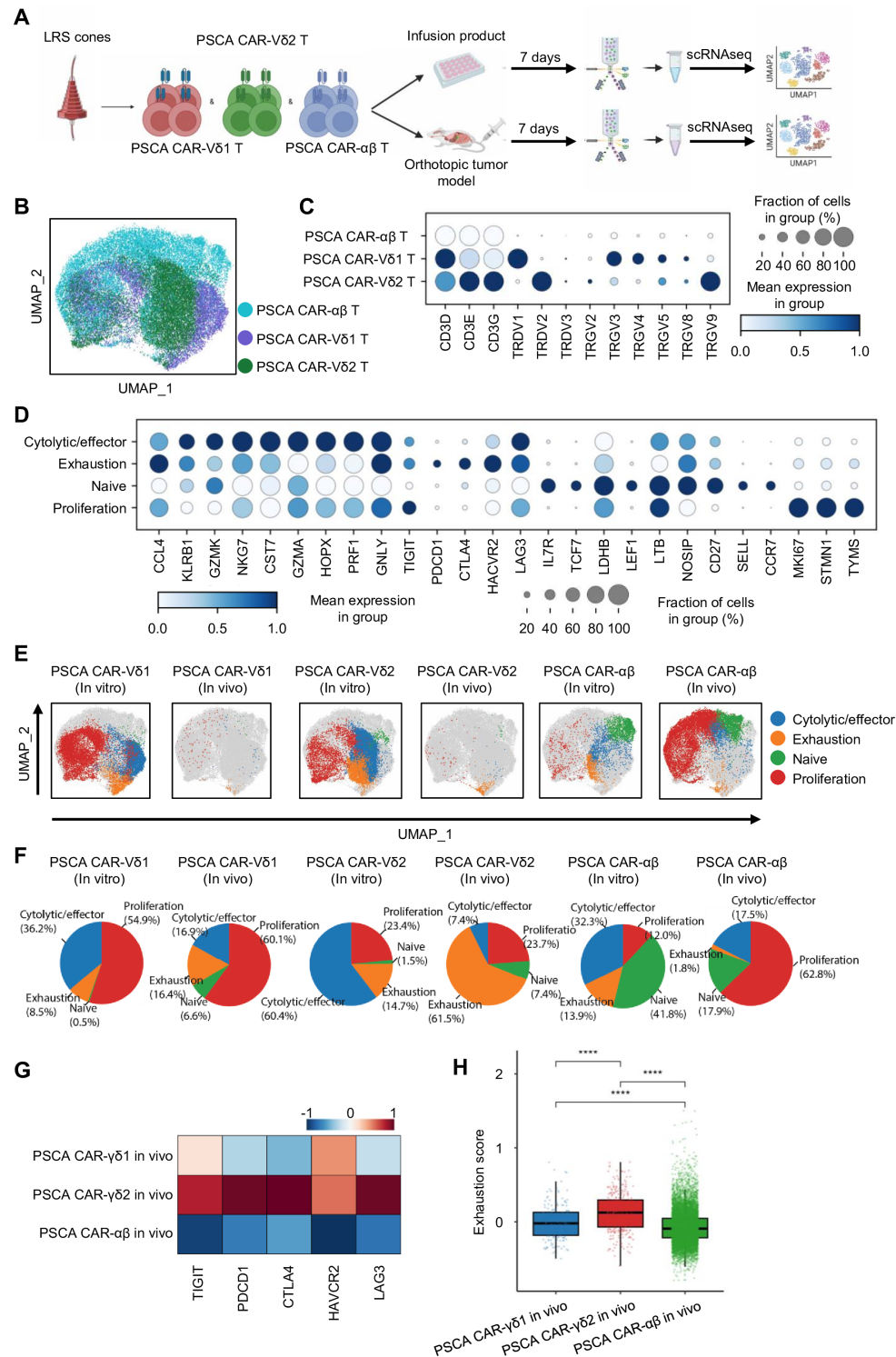


Figure 5 Distinct single-cell transcriptome and functional profile between PSCA CAR-Vδ1, PSCA CAR-Vδ2, and PSCA CAR-αβ T cells. (A) Schematic of the experiment design. (B) Combined UMAP visualizing transcriptional clusters of PSCA CAR-Vδ1, PSCA CAR-Vδ2 T, and PSCA CAR-αβ T cells. (C) Dot plot shows the distinct marker genes for Vδ1, Vδ2, and αβ T cells. (D) Dot plot illustrates functional gene expression across distinct subsets of T cells. (E) UMAP visualizing transcriptional clusters of PSCA CAR-Vδ1, PSCA CAR-Vδ2 T, and PSCA CAR-αβ T cells before (in vitro) and after (in vivo) their infusion into the Capan-1 orthotopic pancreatic cancer model. Cells for each group were pooled from two mice for analysis. (F) Pie charts showing the relative proportions of each of the four transcriptional clusters shown in (E) within PSCA CAR-Vδ1, PSCA CAR-Vδ2, and PSCA CAR-αβ T cells before and after their infusion into the Capan-1 orthotopic pancreatic cancer model. (G) Heatmap displaying exhaustion gene expression profiles in PSCA CAR-Vδ1, CAR-Vδ2, and CAR-αβ T cells in vivo. (H) Box plots of exhaustion scores in PSCA CAR-Vδ1, PSCA CAR-Vδ2, and PSCA CAR-αβ T cells in vivo. The p value was calculated with an unpaired t test, ****p<0.0001. CAR, chimeric antigen receptor; LRS, Leukocyte Reduction System; PSCA, prostate stem cell antigen; scRNA-seq, single-cell RNA sequencing; UMAP, uniform manifold approximation and projection.

the cytotoxic/effector cluster defined by cytotoxic genes including *CCL4*, *KLRB1*, *GZMK*, *NKG7*, *CST7*, *GZMA*, *HOPX*, *PRF1* and *GNLY* underwent a 2.2-fold decrease in expression after adoptive transfer (36.2% in vitro to 16.9% in vivo), possibly resulting from T cell dysfunction that may be induced by the immunosuppressive tumor microenvironment. Although the in vitro PSCA CAR-V δ 2 T cell populations had a higher fraction of cytolytic/effector cells (60.4%) when compared with the in vitro PSCA CAR-V δ 1 T cells (36.2%), the PSCA CAR-V δ 2 T cell population underwent a much more dramatic (8.8-fold) decrease of cytotoxic/effector cluster after adoptive transfer (60.4% in vitro to 7.4% in vivo) (figure 5F). The fraction within the naïve cluster increases when transitioning from in vitro to in vivo for both PSCA CAR-V δ 1 and PSCA CAR-V δ 2 subsets of T cells, yet decreases dramatically in PSCA CAR- $\alpha\beta$ T cells as substantially more cells show a proliferative profile in vivo.⁵² In vivo, both PSCA CAR- $\alpha\beta$ T cells (62.8%), and PSCA CAR-V δ 1 T cells have a much greater fraction of cells in the proliferative cluster (62.8% and 60.1%, respectively), when compared with the PSCA CAR-V δ 2 T cells (~23.7%) (figure 5E and F). These proliferative and phenotypic distinctions underscore the inherent resilience of the PSCA CAR-V δ 1 subset when compared with PSCA CAR-V δ 2 T cells, but also the comparable robustness of PSCA CAR- $\alpha\beta$ T cells. We examined the exhaustion signature (*TIGIT*, *PDCD1*, *CTLA4*, *HAVCR2*, and *LAG3*) following in vivo exposure to tumor. While PSCA CAR- $\alpha\beta$ T cells exhibit the lowest exhaustion score among the three groups, PSCA CAR-V δ 1 T cells exhibited significantly lower exhaustion score compared with PSCA CAR-V δ 2 T cells (figure 5G and H).

Allogeneic PSCA CAR-V δ 1 T cell therapy is free of graft versus host disease and cytokine release syndrome

Graft-versus-host (GvH) responses represent a primary safety concern in allogeneic T cell therapies. However, $\gamma\delta$ T cells are not reactive to mismatched HLA molecules or protein alloantigens and thus are not expected to elicit GvH responses. To assess the safety profile of our PSCA CAR-V δ 1 T cells, we conducted in vivo studies using tumor-bearing NSG mice (figure 6A). No consistent differences in body temperature were observed among the saline, PSCA CAR-V δ 1 T cell, and PSCA CAR- $\alpha\beta$ T cell groups (figure 6B). Systemic GvHD was evaluated using a scoring system encompassing five clinical parameters: weight loss, posture (hunching), activity level, fur texture, and skin integrity.⁵³ Injection of PSCA CAR- $\alpha\beta$ T cells resulted in sustained losses in body weight that were not seen in the mice treated with PSCA CAR-V δ 1 T cells or in the saline-treated controls (figure 6C). Consistent with this, tumor-bearing mice treated with PSCA CAR- $\alpha\beta$ T cells experienced obvious GvHD compared with mice treated with CAR-V δ 1 T cells or saline as evidenced by a significantly higher GvHD score (figure 6D). Indeed, PSCA CAR-V δ 1 T-treated mice maintained a significant survival benefit compared with the mice treated with

PSCA CAR- $\alpha\beta$ T cells who died of GvHD. Likewise, saline-treated mice died of tumor progression (figure 6E).

To assess whether mice treated with PSCA CAR- $\alpha\beta$ T cells suffered inflammatory damage in conventional GvHD target tissues, we performed a histopathological analysis of the colon, liver, and small intestine at the end of the experiment. We found that mice receiving PSCA CAR- $\alpha\beta$ T cells had obvious crypt loss in colonic tissues, inflammatory lymphocyte infiltration in the liver, and apoptotic bodies observed in the small intestine. In contrast, we did not observe such pathological changes following either PSCA CAR-V δ 1 T cell treatment or saline treatment (figure 6F). Additionally, we confirmed that mice treated with V δ 1 T cells in the absence of the PSCA CAR were without evidence of acute GvHD (online supplemental figures 5A–C), indicating the PSCA CAR engineering does not change the safety profiles. Collectively, treatment of mice with human PSCA CAR V δ 1 T cells did not result in an increased risk of GvHD in tumor-bearing NSG mice, compared with mice treated with PSCA CAR- $\alpha\beta$ T cells.

Next, we examined the differences in cytokine release syndrome (CRS) triggered by infusion of PSCA CAR-V δ 1 T versus PSCA CAR- $\alpha\beta$ T cells. Using donor-matched PBMC humanized NSG mice (online supplemental figure 5D), we observed significant fever among mice treated with PSCA CAR- $\alpha\beta$ T cells within 48 hours after infusion, coupled with significantly higher mouse Serum Amyloid A3 and CRS-related IL-6 cytokine release compared with mice receiving PSCA CAR-V δ 1 T cells (online supplemental figure 5E,G and H). Mice treated with either PSCA CAR- $\alpha\beta$ T cells or PSCA CAR-V δ 1 T cells showed a similar body weight change curve (online supplemental figure 5F). Collectively, these results suggest that PSCA CAR-V δ 1 T cells are less likely to induce GvHD and severe CRS, highlighting their promising potential for safe allogeneic applications compared with their conventional PSCA CAR- $\alpha\beta$ T cell counterparts.

DISCUSSION

The development of allogeneic off-the-shelf cell therapies is at the forefront of the rapidly advancing field of cancer immunotherapy. Currently, four major categories of allogeneic cell products are under investigation: $\gamma\delta$ T cells,⁵⁴ engineered conventional $\alpha\beta$ T cells derived from healthy donors,⁵⁵ NK cells,^{43 56} or invariant NKT (iNKT) cells.^{57–59} In the current study, we report the generation and functional characterization of allogeneic PSCA CAR-engineered V δ 1 T cells as a novel approach for managing pancreatic cancer (PC). These off-the-shelf PSCA CAR-V δ 1 T cells exhibit potent cytotoxicity against PC cells in vitro and eradicate established metastatic PC in vivo, while demonstrating a lower risk of GvHD and CRS compared with PSCA CAR- $\alpha\beta$ T cells, and reduced exhaustion relative to PSCA CAR-V δ 2 T cells. This preclinical investigation introduces an allogeneic, off-the-shelf cell therapy candidate for PC with potentially minimal treatment-related

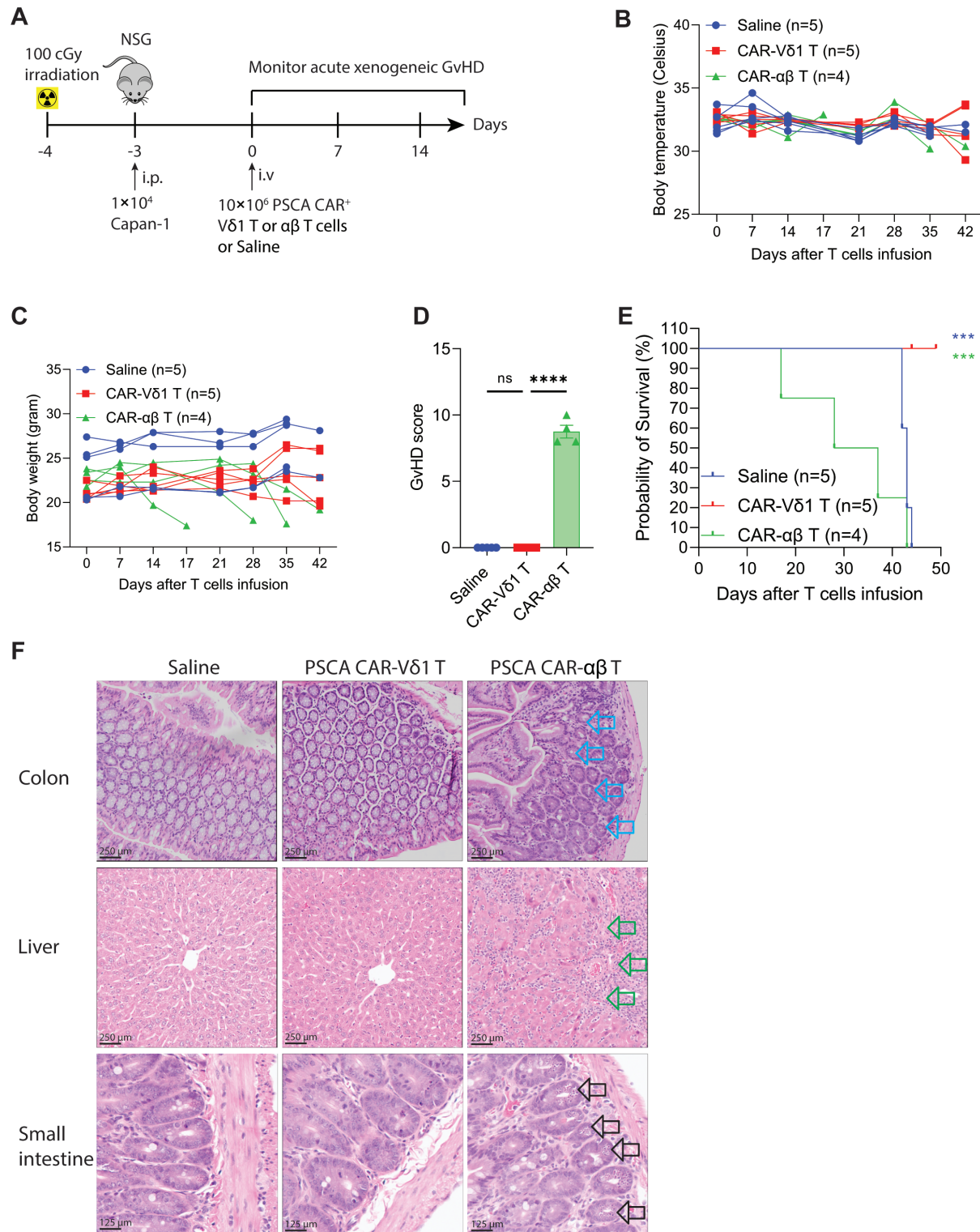


Figure 6 GvHD studies of PSCA CAR-Vδ1 T cells. (A) Experimental design for the graft-versus-host response of PSCA CAR-Vδ1 T cells versus PSCA CAR-αβ T cells using a metastatic PC model. (B) Assessment of body temperature measurement performed on a weekly basis for the mice in (A). (C) Weekly body weight measurement of the mice indicated in (A). (D) GvHD score was measured by two independent researchers at the end of study. The degree of systemic GvHD was assessed by a scoring system described in the Methods' GvHD model section. ****p<0.0001, one-way ANOVA. (E) Kaplan-Meier survival curves of experimental mice over time. ***p<0.001 (log-rank (Mantel-Cox) test). (F) Pathological analysis of major organs collected from different treatment groups when mice were moribund and humanely sacrificed. Representative photomicrographs of H&E-stained sections of colon (10×), liver (10×), and small intestine (20×) are shown. GvHD tissue damage manifests as crypt loss (open arrow in cyan), inflammatory cell infiltration (open arrow in green), and crypt apoptosis (open arrow in black). ANOVA, analysis of variance; CAR, chimeric antigen receptor; GvHD, graft-versus-host disease; i.p., intraperitoneal; i.v., intravenous; ns, not significant; PC, pancreatic cancer; PSCA, prostate stem cell antigen.

toxicity and provides foundational insights to guide the continued advancement of T cell-based immunotherapies for solid tumors.

Conventional cellular immunotherapy with $\alpha\beta$ T cells carries a risk of GvHD in allogeneic settings due to HLA incompatibility. To mitigate this, endogenous TCR expression must be ablated through gene editing.⁶⁰ While this approach enables allogeneic application, it also increases manufacturing complexity.^{61–63} Additionally, despite their robust in vivo clonal expansion, $\alpha\beta$ T cells exhibit limited antitumor efficacy in solid tumors due to poor tissue infiltration. As an alternative, NK cell-based therapies pose a lower risk of GvHD, but their limited in vivo expansion and persistence remain key challenges in improving solid tumor reactivity and therapeutic potential.⁶⁴ Similarly, iNKT cells may offer a promising platform for allogeneic therapy, as they are potent immune effector cells with low GvHD risk and the ability to target tumors with low immunogenicity. Preclinical studies using in vitro-derived iNKT cells from hematopoietic stem cells have demonstrated both strong tumor cytotoxicity and a favorable safety profile.^{57,58}

Engineering innate type $\gamma\delta$ T cells that have potent antitumor capacity while having low to no risk for GvHD represents an exciting direction in the development of allogeneic cell therapy for cancer, especially for solid tumors. Although both V δ 1 and V δ 2 T-cell-based cancer immunotherapies have attracted attention,^{23,65–68} the differences in antitumor potency between these two distinct $\gamma\delta$ T cell subsets have not been extensively explored and remain largely unknown, especially in the context of CAR T cell therapies of solid tumors. While our parallel comparative study provides direct evidence that PSCA CAR V δ 1 T cells and PSCA CAR V δ 2 T cells have antitumor efficacy against PC comparable to that of conventional PSCA CAR $\alpha\beta$ T cells, further investigation is necessary to fully determine which subset may prove superior and practical for therapeutic applications in clinic. Such work includes additional preclinical and clinical comparisons between PSCA CAR V δ 1 T cells and PSCA CAR V δ 2 T cells in the setting of PC and other PSCA⁺ solid tumors that will provide important additional insights into the field of $\gamma\delta$ T cell-based allogeneic immunotherapy.

Unlike V δ 2 T cells, which recognize butyrophilin complex 3A1 through a unique mechanism involving intracellular phosphoantigen sensing,⁶⁹ the activation mechanism of V δ 1 T cells remains incompletely understood. Members of the CD1 family of MHC-like proteins specifically the lipid-presenting molecules CD1c and CD1d have been identified as ligands for certain V δ 1 TCRs; however, their capacity to elicit V δ 1 T cell responses in vivo is still uncertain.^{70–72} Ongoing ligand discovery and characterization efforts aim to define the specific triggers of clonally expanded V δ 1 T cells, which could further enable their clinical application. In addition to the TCR, PSCA CAR V δ 1 T cells also express innate NK cell receptors. For instance, activation of NKG2D by MICA/B and ULBP ligands promotes GZMB and perforin release,^{73,74}

while DNAM-1 mediates cytotoxicity through recognition of CD155 and CD112, which are broadly expressed on both hematologic and solid tumors.⁷⁵ However, the data presented in this report indicate that their contribution, relative to that of the PSCA CAR, is minimal at best.

In summary, we performed a head-on preclinical comparison of allogeneic PSCA CAR V δ 1 T cells and PSCA CAR V δ 2 T cells and PSCA CAR $\alpha\beta$ T cells in the context of PC, a devastating malignancy that has not had significant therapeutic advances in 40 years. Given the metastatic characteristics of the Capan-1 cell line used in our in vivo model, we adopted the combined intravenous and i.p. infusion strategy. The rationale for using both intravenous and i.p. routes for effector T cell and mock T cell infusion is based on our previously published study where we demonstrated that intravenously injected NK cells primarily accumulated in the liver and lungs, whereas i.p.-injected NK cells localized mainly to the liver and pancreas.⁴³ Notably, we also employed this dual-route approach for PSCA CAR-macrophage therapy in a recent publication, which showed promising in vivo tumor-suppressive effects.⁴⁶ Based on this collective evidence, we hypothesize that the intravenous and i.p. combination may similarly enhance the antitumor efficacy of $\gamma\delta$ T cells. We found that while all three T cell types showed equal and rigorous antitumor activity against PC in vivo, PSCA CAR V δ 1 T cells appeared to be less toxic compared with PSCA CAR $\alpha\beta$ T cells as measured by GvHD and CRS, and possibly more robust compared with PSCA CAR V δ 2 T cells as measured by scRNA-seq, although this must be validated in vivo in future studies. While significant progress has been made in expanding V δ 1 T cells, achieving reliable and scalable production of CAR-V δ 1 T cells in clinically relevant quantities continues to be a challenge. Exploring advanced enrichment methods and innovative approaches, such as the generation of iPSC-derived CAR-V δ 1 T cells, could provide critical insights and address current limitations, paving the way for more efficient and universally applicable therapies. Addressing these challenges will be essential for translating this platform into a robust and scalable clinical solution for cancer immunotherapy.

MATERIALS AND METHODS

Cell lines

The human PC cell lines Capan-1, MIA PaCa-2, and PANC-1 were obtained from American Type Culture Collection (ATCC) and cultured in Dulbecco's Modified Eagle's Medium (DMEM) supplemented with 10% fetal bovine serum (FBS), penicillin (100 U/mL), and streptomycin (100 μ g/mL). No additional authentication was performed, as all cell lines were sourced directly from ATCC. Capan-1 cells stably expressing the luciferase_ZsGreen reporter gene were generated via lentiviral transduction with a vector encoding Luc_ZsGreen. Early-passage cells were cryopreserved, and prior to experiments, they were thawed and passaged two to three times

to achieve viability greater than 95%. All cell lines were routinely screened for mycoplasma contamination using the MycoAlert Plus Mycoplasma Detection Kit (Lonza).

DNA constructs and retrovirus production

The CAR targeting PSCA was cloned into a clinical-grade retroviral replicating vector (RRV), from which non-human DNA sequences were minimized, as previously described.⁴³ The PSCA CAR was designed to sequentially include a signal peptide, anti-PSCA single-chain fragment variable (scFv), hinge region, transmembrane domain, CD28, and CD3z. The anti-PSCA scFv-hinge-CD28-CD3z-T2A construct was inserted into the RRV at the NotI/SalI restriction sites. A tEGFR, serving both as a traceable marker and in vivo suicide switch, was linked in-frame to the PSCA CAR via a porcine teschovirus-1 2A (P2A)-like self-cleaving peptide⁴³ and cloned into the RRV at the SalI/MluI site. Control plasmids expressing only tEGFR were generated by PCR amplification of the relevant sequence from the PSCA CAR-P2A-tEGFR construct and cloned into the retroviral vector at the NotI/MluI sites. Retrovirus expressing PSCA CAR-P2A-tEGFR was produced by cotransfecting the transfer plasmid with plasmid pRD114-TR, which encodes the envelope glycoprotein, into GP2-293T cells (Takara, San Francisco, California, USA) using Lipofectamine 3000 (Thermo Fisher, Carlsbad, California, USA). Supernatants were collected 48 hours post-transfection, filtered through a 0.45 µm membrane, and cryopreserved in liquid nitrogen for future use.

T cell isolation, expansion, and transduction

PBMCs were isolated from whole blood leukocyte cones using Ficoll-Paque PLUS (Cytiva) density gradient centrifugation according to the manufacturer's instructions. Freshly isolated PBMCs were either cryopreserved in 90% FBS with 10% dimethyl sulfoxide (DMSO) for future use or resuspended at a density of 4×10^6 cells/mL in Vδ1 T cell expansion medium. This medium consisted of Roswell Park Memorial Institute (RPMI) 1640 supplemented with 10% FBS (v/v), 1% penicillin/streptomycin, 1% MEM NEAA, 10 mM HEPES, 1 mM sodium pyruvate, 50 µM β-mercaptoethanol (β-ME), 2.5 µg/mL Plasmocin Prophylactic, 1 µg/mL phytohemagglutinin (PHA), 1 µg/mL Concanavalin A, 100 IU/mL IL-2, 20 ng/mL IL-7, 10 ng/mL IL-15, 10 ng/mL IL-21, and 10 ng/mL IFN-γ.

An additional PBMC aliquot was resuspended in Vδ2 T cell expansion medium, composed of RPMI 1640 with 10% FBS (v/v), 1% penicillin/streptomycin, 1% MEM NEAA, 10 mM HEPES, 1 mM sodium pyruvate, 50 µM β-ME, 2.5 µg/mL Plasmocin Prophylactic, 200 IU/mL IL-2, and 5 µM zoledronic acid monohydrate, also at 4×10^6 cells/mL. For αβ T cell expansion, PBMCs were resuspended in RPMI 1640 medium supplemented with 10% FBS (v/v), 1% penicillin/streptomycin, 1% MEM NEAA, 10 mM HEPES, 1 mM sodium pyruvate, 50 µM

β-ME, 2.5 µg/mL Plasmocin Prophylactic, and 200 IU/mL IL-2.

When cryopreserved PBMCs were used, they were thawed and rested overnight in prewarmed complete media at 4×10^6 cells/mL prior to further processing to minimize lymphocyte stress. After 14–21 days of γδ T cell enrichment, Vδ1 and Vδ2 cells were sorted by flow cytometry based on the gating strategy hCD45⁺hCD3⁺TCR Vδ1⁺ or hCD45⁺hCD3⁺TCR Vδ2⁺, respectively. αβ T cells were isolated on the day of PBMC collection using hCD45⁺hCD3⁺TCR αβ⁺ gating. Sorted Vδ1 and Vδ2 γδ T cells were cultured at 1×10^6 cells/mL in the optimized expansion medium described above. Irradiated K562 feeder cells were added weekly at a 1:100 ratio (γδ T:K562) to promote expansion. Culture medium was replaced every 2–3 days, and cells were washed with phosphate-buffered saline as needed. All cells were routinely tested for mycoplasma contamination.

Before PSCA CAR genetic modification, 1 million Vδ1 γδ, Vδ2 γδ, or αβ T cells were activated with 25 µL CD3/CD28 activator beads in 1 mL of culture medium for 48 hours. Cells were then transduced with either the PSCA retrovirus or the tEGFR control virus using Retrofectin (Takara) according to the manufacturer's protocol and expanded until use. Transduction efficiency was assessed by flow cytometric analysis of eGFR expression 48 hours post-transduction. Viable cells were counted by Trypan blue exclusion using a hemocytometer. Finally, transduced T cells were cryopreserved in 90% FBS+10% DMSO, and all functional assays were performed using thawed cells after recovery.

Annexin V/DAPI staining killing assay

To evaluate T cell-mediated tumor cell death, GFP⁺ Capan-1 tumor cells (5,000 cells per well) were co-cultured for 24 hours with either mock or PSCA CAR-engineered Vδ1, Vδ2, or αβ T effector cells at varying E:T ratios (1:2, 1:1, 2:1; n=3 donors). After co-culture, cells were stained with APC-conjugated Annexin V and 4',6-diamidino-2-phenylindole (DAPI) (BD Biosciences), and analyzed by flow cytometry. For intracellular cytokine staining, mock or PSCA CAR-engineered Vδ1, Vδ2, or αβ T effector cells were co-cultured with tumor cells in the presence of BD GolgiPlug (BD Biosciences) at 37°C for 4 hours. Following co-culture, cells underwent surface staining, fixation, and permeabilization using the Cytofix/Cytoperm Fixation/Permeabilization Solution Kit (BD Biosciences). Intracellular staining for GZMB and IFNγ was then performed according to the manufacturer's protocol.

⁵¹Cr-based cell killing assay

Cytotoxicity assays were conducted as previously described. Target cells including Capan-1, MIA PaCa-2, and PANC-1 were labeled with ⁵¹Cr for 90 min at 37°C, then washed and co-incubated with effector cells (Vδ1 γδ cells, Vδ2 γδ cells, or αβ T cells) in 96-well V-bottom plates at varying E:T ratios for 24 hours at 37°C. Following

incubation, supernatants were collected into scintillation vials for analysis. The standard formula of $100 \times (\text{cpm}_{\text{experimental}} - \text{cpm}_{\text{spontaneous}}) / (\text{cpm}_{\text{maximal}} - \text{cpm}_{\text{spontaneous}})$ was used to calculate percentages of specific lysis, which were displayed as the means of triplicate samples.

xCELLigence real-time cytotoxicity assay

The cytolytic activity of Vδ1 T cells was assessed using the xCELLigence RTCA Multiple Plates instrument (ACEA Biosciences), as previously described.^{47–49} E-plates were calibrated with 100 μL of DMEM supplemented with 10% FBS for 10 min. Subsequently, 10,000 PSCA-expressing Capan-1 cells in 50 μL of DMEM with 10% FBS were seeded per well and allowed to adhere for 24 hours at 37°C in a 5% CO₂ atmosphere. Next, 5,000 CAR-expressing Vδ1 T cells were added at 1:2 E:T ratio in 50 μL Vδ1 T cell medium (RPMI1640 supplemented with 10% FBS (V/V), 1% Penicillin/Streptomycin, 1% MEM NEAA, 10 mM HEPES, 1 mM sodium pyruvate, 50 μM β-ME, 2.5 μg/mL Plasmocin Prophylactic, 1 μg/mL PHA, 1 μg/mL Concanavalin A, 100 IU/mL IL-2, 20 ng/mL IL-7, 10 ng/mL IL-15, 10 ng/mL IL-21, 10 ng/mL IFN-γ) per well to bring the final volume to 200 μL 24 hours after Capan-1 cell seeding. Tumor cell viability is measured as Tumor Cell Index (impedance per well),^{47–49} and was recorded every 15 min during tumor cell growth. Once the Tumor Cell Index decreased nearing zero, indicating that adherent Capan-1 cells inducing impedance have been cleared by PSCA CAR-Vδ1 T cells, another 10,000 PSCA-expressing Capan-1 cells were added at two separate intervals over the following 7 days. Thus, by the end of the assay, a total of 30,000 tumor cells had been exposed to 5,000 activated PSCA CAR-Vδ1 T cells. Six different donors were assessed, and each condition was run in duplicate wells.

Pancreatic cancer mouse model

NOD.Cg-PrkdcSCIDIl2rgtm1Wjl/SzJ (NOD/SCID/IL-2Rγ^{-/-}, NSG) mice were housed in the animal facilities at the City of Hope. Male and female mice aged 6–10 weeks (The Jackson Laboratory) were used for all experiments unless otherwise specified. On day 4, immunodeficient NSG mice were engrafted i.p. with 5 × 10⁵ Capan-1 cells expressing the luciferase_ZsGreen gene. Mice were then randomly assigned to vehicle or treatment groups. PSCA CAR-Vδ1 T cells or mock T cells (which are T cells transduced with the tEGFR without the PSCA CAR) were thawed from liquid nitrogen and administered via both i.p. and intravenous routes. For i.p. dosing, mice received either a high dose (4.5 × 10⁶ CAR⁺ Vδ1 T cells) or low dose (2.25 × 10⁶ CAR⁺ Vδ1 T cells); for intravenous dosing, high and low doses were 1.5 × 10⁶ and 0.75 × 10⁶ CAR⁺ Vδ1 T cells, respectively. A total of three doses of PSCA CAR-Vδ1 T cells, mock Vδ1 T cells, or vehicle (saline) were given on day 0, day 14, and day 31. All animal experiments were conducted in accordance with Animal Research Reporting In Vivo Experiments, federal, state, and local guidelines with approval from the City of Hope Institutional Animal Care and Use Committee.

Single-cell RNA sequencing library preparation

The scRNA-seq libraries were prepared using the 10x Genomics Chromium Single Cell 5' Library & Gel Bead Kit v2 (Chromium Single Cell 5' Library & Gel Bead Kit v2, Cat# PN-1000265). Single cells were isolated from a cell suspension, ensuring they had suitable viability and concentration for single-cell capture. These cells were then captured and barcoded using the 10x Genomics Chromium Controller, where each cell was encapsulated into Gel Beads-in-Emulsion (GEMs) containing barcoded oligonucleotides. Within the GEMs, cell lysis and reverse transcription occurred, producing barcoded full-length complementary DNA (cDNA), which was subsequently purified and amplified via PCR. The amplified cDNA underwent library construction, including fragmentation, end repair, A-tailing, adaptor ligation, and index PCR to add unique dual indices for multiplexing. Quality and quantity assessments of the cDNA and final libraries were performed using an Agilent 2100 Bioanalyzer and a Qubit Fluorometer to ensure they met sequencing requirements. The libraries were sequenced on an Illumina platform with a configuration of 151 bp paired-end reads and an additional 11 bp for sample indexing (151+11+11+151), providing comprehensive and high-quality data for downstream analysis.

Single-cell RNA-seq data analysis

Raw sequencing data from each sample were aligned to the human reference genome (GRCh38) using the Cell Ranger count command, generating single-cell resolution expression data following the 10x Genomics protocol (<https://support.10xgenomics.com/single-cell-gene-expression/software/pipelines/latest/using/count>). Downstream analysis was conducted using Scanpy.⁷⁶ Cell quality was assessed based on two metrics: (1) the number of detected genes and total gene counts per cell, and (2) the proportion of mitochondrial and hemoglobin gene counts per cell. Cells were excluded if they had fewer than 300 detected genes, fewer than 300 or more than 7,000 total gene counts, mitochondrial UMI content exceeding 20%, or hemoglobin gene expression greater than 3%. Genes detected in fewer than four cells were also filtered out.

Subsequently, raw counts were normalized using Scanpy's `normalize_total` function (with `target_sum=1e4`), followed by natural log transformation after adding a pseudocount of 1 to stabilize variance. Highly variable genes (HVGs) were identified within each sample using `scanpy.pp.highly_variable_genes` with the parameter `batch_key="samples"`. To account for unwanted variation, total counts and mitochondrial gene percentages were regressed out using `scanpy.pp.regress_out`, applied to the HVG expression matrix.

Principal component analysis (PCA) was performed using `scanpy.tl.pca`, and the top 50 components were retained for further analysis. Batch effects were corrected using the Batch Balanced k-Nearest Neighbors (BBKNN) algorithm,⁷⁶ which identifies each cell's nearest neighbors

Table 1 Assessment of clinical GvHD in transplanted animals⁵³

Criteria	Grade 0	Grade 1	Grade 2
Weight loss	<10%	>10% to <20%	>20%
Posture	Normal	Hunching noted only at rest	Severe hunching impairs movement
Activity	Normal	Mild-to-moderately decreased	Stationary unless stimulated
Fur texture	Normal	Mild-to-moderate ruffling	Severe ruffling/poor grooming
Skin integrity	Normal	Scaling of paws/tail	Obvious areas of denuded skin

GvHD, graft-versus-host disease.

within each batch rather than across the pooled dataset. BBKNN was executed via `scanpy.external.pp.bbkn` with parameters `batch_key="sample"` and `n_pcs=50`. UMAP for dimensionality reduction was then applied using `scanpy.tl.umap` with default settings.

To identify Vδ subsets in the scRNA-seq dataset, we implemented the multimodal classifier hierarchical classification (MMoCHi) strategy,⁷⁷ which is trained using events defined by thresholding the expression of user-specified marker genes. Specifically, *TRDV1* and *TRDV2* expression levels were used to annotate Vδ1 and Vδ2 subsets, respectively, thereby excluding contaminating cell populations.

GvHD model

GvHD severity was evaluated by monitoring body weight changes, a well-established indicator of systemic GvHD in this and other murine models.^{53, 77} Transplanted mice were ear-punched for identification, and individual body weights were recorded on day 0 and weekly thereafter until analysis. A weight loss exceeding 10% was used as the threshold for moderate GvHD, based on prior reports identifying this level as indicative of significant disease. Systemic GvHD was further assessed using a clinical scoring system outlined in [table 1](#), which incorporates five parameters: weight loss, posture (hunching), activity, fur texture, and skin integrity. At the time of analysis, mice from coded cages were scored from 0 to 2 for each parameter, and a clinical index was calculated by summing the scores (maximum possible score=10).

Overall GvHD severity was determined by combining objective histopathological assessments of target organs (skin, liver, small intestine, and colon) with subjective performance evaluations by two independent researchers.

Time-lapse bioluminescence live animal imaging

Time-lapse BLI was performed using a Lago-X (Spectral Instruments Imaging) following the manufacturer's instructions. The mice were anesthetized with 2-4% isoflurane and oxygen (1L/min) in an imaging chamber. Live animal imaging was acquired 5 min after S.C. injection of D-Luciferin (3 mg/mouse, Goldbio) prepared under the manufacturer's

instructions. Luminescence images were quantified by Aura Imaging Software (V.2.2.1.1).

Immunohistochemistry and H&E assay

Tissues including pancreas, liver, lung, tumor, and spleen were harvested from experimental mice at the time of euthanasia, carefully trimmed of fat and surrounding connective tissue, and fixed in 10% neutral buffered formalin for up to 36 hours. Fixed samples were subsequently embedded in paraffin and sectioned at a thickness of 4-6 μm. Tissue sections were stained with H&E by the City of Hope Pathology Core in accordance with standard protocols. Stained slides were imaged at 20× and 40× magnifications using an Olympus BX51 upright microscope equipped with an Optronics Macrofire CCD camera (AU Optronics), and images were analyzed using Optronics Picture-Frame software (AU Optronics).

Statistical analysis

For continuous endpoints assuming normal or symmetric distributions, data are presented as mean±SEM for each group. The Student's t-test was used to compare two independent groups. The one-way analysis of variance model was used to compare three or more independent groups. For survival data, survival functions are estimated using the Kaplan-Meier method and compared using the log-rank test. All tests were two-sided, and p values were adjusted for multiple comparisons using Holm's procedure. A p value<0.05 was defined as statistically significant. GraphPad V.9.1.0 and SAS V.9.4 were used for statistical analysis in the current study. Flow cytometry, ELISA, and BLI are listed in the online supplemental Materials.

Author affiliations

¹Department of Hematology & Hematopoietic Cell Transplantation, City of Hope National Medical Center, Los Angeles, California, USA

²Hematologic Malignancies Research Institute, City of Hope National Medical Center, Los Angeles, California, USA

³Department of Computational and Quantitative Medicine, City of Hope National Medical Center, Los Angeles, California, USA

⁴City of Hope Comprehensive Cancer Center, Los Angeles, California, USA

⁵Division of Hematology & Oncology, Department of Medicine, School of Medicine, University of California, Irvine, California, USA

⁶Institute for Precision Cancer Therapeutics and Immuno-Oncology, Chao Family Comprehensive Cancer Center, University of California, Irvine, California, USA

⁷The Clemons Family Center for Transformative Cancer Research, University of California, Irvine, California, USA

Acknowledgements We thank the City of Hope's Analytical Cytometry Core and Pathology Core.

Contributors Conceptualization: JY, MAC. Methodology: Zhixin Li, NL, ZS, LJ, LT, XS, JZ, Zhiyao Li. Investigation: Zhixin Li, NL, ZS, LJ, Visualization: Zhixin Li, NL, ZS, Funding acquisition: JY, MAC. Project administration: JY, MAC. Supervision: JY, MAC. Writing, review and editing: Zhixin Li, ZS, LJ, LT, MAC, JY. Guarantors: MAC, JY.

Funding National Institutes of Health grant CA265095 (MAC) National Institutes of Health grant CA210087 (MAC) National Institutes of Health grant CA264512 (JY) National Institutes of Health grant CA266457 (JY) National Institutes of Health grant CA163205 (MAC).

Competing interests No, there are no competing interests.

Patient consent for publication Not applicable.

Ethics approval All experiments and animal procedures were conducted in accordance with federal, state, and local regulations and were approved by the City of Hope Institutional Animal Care and Use Committee (ID:21062/233320). Peripheral blood cones were obtained from healthy donors with written informed consent under a protocol approved by the City of Hope Institutional Review Board.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available upon reasonable request. All data are available in the main text or the supplementary materials. Raw data generated in this study are available upon reasonable request from the corresponding authors.

Supplemental material This content has been supplied by the author(s). It has not been vetted by BMJ Publishing Group Limited (BMJ) and may not have been peer-reviewed. Any opinions or recommendations discussed are solely those of the author(s) and are not endorsed by BMJ. BMJ disclaims all liability and responsibility arising from any reliance placed on the content. Where the content includes any translated material, BMJ does not warrant the accuracy and reliability of the translations (including but not limited to local regulations, clinical guidelines, terminology, drug names and drug dosages), and is not responsible for any error and/or omissions arising from translation and adaptation or otherwise.

Open access This is an open access article distributed in accordance with the Creative Commons Attribution Non Commercial (CC BY-NC 4.0) license, which permits others to distribute, remix, adapt, build upon this work non-commercially, and license their derivative works on different terms, provided the original work is properly cited, appropriate credit is given, any changes made indicated, and the use is non-commercial. See <http://creativecommons.org/licenses/by-nc/4.0/>.

ORCID iD

Michael A Caligiuri <http://orcid.org/0000-0002-4095-3020>

REFERENCES

- 1 Siegel RL, Miller KD, Wagle NS, *et al.* Cancer statistics, 2023. *CA Cancer J Clin* 2023;73:17–48.
- 2 Rahib L, Smith BD, Aizenberg R, *et al.* Projecting cancer incidence and deaths to 2030: the unexpected burden of thyroid, liver, and pancreas cancers in the United States. *Cancer Res* 2014;74:2913–21.
- 3 Burris HA 3rd, Moore MJ, Andersen J, *et al.* Improvements in survival and clinical benefit with gemcitabine as first-line therapy for patients with advanced pancreas cancer: a randomized trial. *J Clin Oncol* 1997;15:2403–13.
- 4 Amrutkar M, Gladhaug IP. Pancreatic Cancer Chemoresistance to Gemcitabine. *Cancers (Basel)* 2017;9:157.
- 5 Henriksen A, Dyhl-Polk A, Chen I, *et al.* Checkpoint inhibitors in pancreatic cancer. *Cancer Treat Rev* 2019;78:17–30.
- 6 Groh V, Porcelli S, Fabbri M, *et al.* Human lymphocytes bearing T cell receptor gamma/delta are phenotypically diverse and evenly distributed throughout the lymphoid system. *J Exp Med* 1989;169:1277–94.
- 7 Carding SR, Egan PJ. Gammadelta T cells: functional plasticity and heterogeneity. *Nat Rev Immunol* 2002;2:336–45.
- 8 Gentles AJ, Newman AM, Liu CL, *et al.* The prognostic landscape of genes and infiltrating immune cells across human cancers. *Nat Med* 2015;21:938–45.
- 9 Wu Y, Biswas D, Usaite I, *et al.* A local human Vδ1 T cell population is associated with survival in nonsmall-cell lung cancer. *Nat Cancer* 2022;3:696–709.
- 10 Wilhelm M, Kunzmann V, Eckstein S, *et al.* Gammadelta T cells for immune therapy of patients with lymphoid malignancies. *Blood* 2003;102:200–6.
- 11 Dieli F, Vermijlen D, Fulfaro F, *et al.* Targeting human {gamma}delta T cells with zoledronate and interleukin-2 for immunotherapy of hormone-refractory prostate cancer. *Cancer Res* 2007;67:7450–7.
- 12 Meraviglia S, Eberl M, Vermijlen D, *et al.* In vivo manipulation of Vgamma9Vdelta2 T cells with zoledronate and low-dose interleukin-2 for immunotherapy of advanced breast cancer patients. *Clin Exp Immunol* 2010;161:290–7.
- 13 Kobayashi H, Tanaka Y, Yagi J, *et al.* Safety profile and anti-tumor effects of adoptive immunotherapy using gamma-delta T cells against advanced renal cell carcinoma: a pilot study. *Cancer Immunol Immunother* 2007;56:469–76.
- 14 Wada I, Matsushita H, Noji S, *et al.* Intraperitoneal injection of in vitro expanded Vγ9Vδ2 T cells together with zoledronate for the treatment of malignant ascites due to gastric cancer. *Cancer Med* 2014;3:362–75.
- 15 Nicol AJ, Tokuyama H, Mattarollo SR, *et al.* Clinical evaluation of autologous gamma delta T cell-based immunotherapy for metastatic solid tumours. *Br J Cancer* 2011;105:778–86.
- 16 Izumi T, Kondo M, Takahashi T, *et al.* Ex vivo characterization of γδ T-cell repertoire in patients after adoptive transfer of Vγ9Vδ2 T cells expressing the interleukin-2 receptor β-chain and the common γ-chain. *Cytotherapy* 2013;15:481–91.
- 17 Kakimi K, Matsushita H, Murakawa T, *et al.* γδ T cell therapy for the treatment of non-small cell lung cancer. *Transl Lung Cancer Res* 2014;3:23–33.
- 18 Khan AN, Asija S, Pendhari J, *et al.* CAR-T cell therapy in hematological malignancies: Where are we now and where are we heading for? *Eur J Haematol* 2024;112:6–18.
- 19 Giorgioni L, Ambrosone A, Cometa MF, *et al.* CAR-T State of the Art and Future Challenges, A Regulatory Perspective. *Int J Mol Sci* 2023;24:11803.
- 20 Deniger DC, Switzer K, Mi T, *et al.* Bispecific T-cells expressing polyclonal repertoire of endogenous γδ T-cell receptors and introduced CD19-specific chimeric antigen receptor. *Mol Ther* 2013;21:638–47.
- 21 Ang WX, Ng YY, Xiao L, *et al.* Electroporation of NKG2D RNA CAR Improves Vγ9Vδ2 T Cell Responses against Human Solid Tumor Xenografts. *Mol Ther Oncolytics* 2020;17:421–30.
- 22 Rozenbaum M, Meir A, Aharon Y, *et al.* Gamma-Delta CAR-T Cells Show CAR-Directed and Independent Activity Against Leukemia. *Front Immunol* 2020;11:1347.
- 23 Makkouk A, Yang XC, Barca T, *et al.* Off-the-shelf Vδ1 gamma delta T cells engineered with glypican-3 (GPC-3)-specific chimeric antigen receptor (CAR) and soluble IL-15 display robust antitumor efficacy against hepatocellular carcinoma. *J Immunother Cancer* 2021;9:e003441.
- 24 Frieling JS, Tordesillas L, Bustos XE, *et al.* γδ-Enriched CAR-T cell therapy for bone metastatic castrate-resistant prostate cancer. *Sci Adv* 2023;9:eadf0108.
- 25 Lin M, Zhang X, Liang S, *et al.* Irreversible electroporation plus allogenic Vγ9Vδ2 T cells enhances antitumor effect for locally advanced pancreatic cancer patients. *Signal Transduct Target Ther* 2020;5:215.
- 26 Oberg H-H, Peipp M, Kellner C, *et al.* Novel bispecific antibodies increase γδ T-cell cytotoxicity against pancreatic cancer cells. *Cancer Res* 2014;74:1349–60.
- 27 Rhodes DA, Chen H-C, Price AJ, *et al.* Activation of human γδ T cells by cytosolic interactions of BTN3A1 with soluble phosphoantigens and the cytoskeletal adaptor periplakin. *J Immunol* 2015;194:2390–8.
- 28 Hintz M, Reichenberg A, Altincicek B, *et al.* Identification of (E)-4-hydroxy-3-methyl-but-2-enyl pyrophosphate as a major activator for human gammadelta T cells in Escherichia coli. *FEBS Lett* 2001;509:317–22.
- 29 Tanaka Y, Morita CT, Tanaka Y, *et al.* Natural and synthetic non-peptide antigens recognized by human gamma delta T cells. *Nature New Biol* 1995;375:155–8.
- 30 Xiao L, Chen C, Li Z, *et al.* Large-scale expansion of Vγ9Vδ2 T cells with engineered K562 feeder cells in G-Rex vessels and their use as chimeric antigen receptor-modified effector cells. *Cytotherapy* 2018;20:420–35.
- 31 Tanaka Y, Murata-Hirai K, Iwasaki M, *et al.* Expansion of human γδ T cells for adoptive immunotherapy using a bisphosphonate prodrug. *Cancer Sci* 2018;109:587–99.
- 32 Almeida AR, Correia DV, Fernandes-Platzgummer A, *et al.* Delta One T Cells for Immunotherapy of Chronic Lymphocytic Leukemia:

- Clinical-Grade Expansion/Differentiation and Preclinical Proof of Concept. *Clin Cancer Res* 2016;22:5795–804.
- 33 Ferry GM, Agbuduwe C, Forrester M, et al. A Simple and Robust Single-Step Method for CAR-V δ 1 $\gamma\delta$ T Cell Expansion and Transduction for Cancer Immunotherapy. *Front Immunol* 2022;13:863155.
 - 34 Simões AE, Di Lorenzo B, Silva-Santos B. Molecular Determinants of Target Cell Recognition by Human $\gamma\delta$ T Cells. *Front Immunol* 2018;9:929.
 - 35 Correia DV, Fogli M, Hudspeth K, et al. Differentiation of human peripheral blood V δ 1+ T cells expressing the natural cytotoxicity receptor Nkp30 for recognition of lymphoid leukemia cells. *Blood* 2011;118:992–1001.
 - 36 Rancan C, Arias-Badia M, Dogra P, et al. Exhausted intratumoral V δ 2⁺ $\gamma\delta$ T cells in human kidney cancer retain effector function. *Nat Immunol* 2023;24:612–24.
 - 37 Davies D, Kamdar S, Woolf R, et al. PD-1 defines a distinct, functional, tissue-adapted state in V δ 1⁺ T cells with implications for cancer immunotherapy. *Nat Cancer* 2024;5:420–32.
 - 38 Li E, Liu L, Li F, et al. PSCA promotes prostate cancer proliferation and cell-cycle progression by up-regulating c-Myc. *Prostate* 2017;77:1563–72.
 - 39 Gu Z, Thomas G, Yamashiro J, et al. Prostate stem cell antigen (PSCA) expression increases with high gleason score, advanced stage and bone metastasis in prostate cancer. *Oncogene* 2000;19:1288–96.
 - 40 Wenthe MN, Jain A, Kono E, et al. Prostate stem cell antigen is a putative target for immunotherapy in pancreatic cancer. *Pancreas* 2005;31:119–25.
 - 41 Maitra A, Adsay NV, Argani P, et al. Multicomponent analysis of the pancreatic adenocarcinoma progression model using a pancreatic intraepithelial neoplasia tissue microarray. *Mod Pathol* 2003;16:902–12.
 - 42 Dorff TB, Blanchard S, Martirosyan H, et al. Final results from phase I study of PSCA-targeted chimeric antigen receptor (CAR) T cells in patients with metastatic castration resistant prostate cancer (mCRPC). *JCO* 2023;41:5019.
 - 43 Teng K-Y, Mansour AG, Zhu Z, et al. Off-the-Shelf Prostate Stem Cell Antigen-Directed Chimeric Antigen Receptor Natural Killer Cell Therapy to Treat Pancreatic Cancer. *Gastroenterology* 2022;162:1319–33.
 - 44 Leong JW, Chase JM, Romee R, et al. Preactivation with IL-12, IL-15, and IL-18 induces CD25 and a functional high-affinity IL-2 receptor on human cytokine-induced memory-like natural killer cells. *Biol Blood Marrow Transplant* 2014;20:463–73.
 - 45 Deng W, Gowen BG, Zhang L, et al. Antitumor immunity. A shed NKG2D ligand that promotes natural killer cell activation and tumor rejection. *Science* 2015;348:136–9.
 - 46 Shah Z, Tian L, Li Z, et al. Human anti-PSCA CAR macrophages possess potent antitumor activity against pancreatic cancer. *Cell Stem Cell* 2024;31:803–17.
 - 47 Cerignoli F, Abassi YA, Lamarche BJ, et al. In vitro immunotherapy potency assays using real-time cell analysis. *PLoS One* 2018;13:e0193498.
 - 48 Zhu J, Wang X, Xu X, et al. Dynamic and label-free monitoring of natural killer cell cytotoxic activity using electronic cell sensor arrays. *J Immunol Methods* 2006;309:25–33.
 - 49 Oberg HH, Peters C, Kabelitz D, et al. Real-time cell analysis (RTCA) to measure killer cell activity against adherent tumor cells in vitro. *Methods Enzymol* 2020;631:429–41.
 - 50 Stefanowicz-Hajduk J, Ochocka JR. Real-time cell analysis system in cytotoxicity applications: Usefulness and comparison with tetrazolium salt assays. *Toxicol Rep* 2020;7:335–44.
 - 51 Becht E, McInnes L, Healy J, et al. Dimensionality reduction for visualizing single-cell data using UMAP. *Nat Biotechnol* 2018.
 - 52 Gray JI, Caron DP, Wells SB, et al. Human $\gamma\delta$ T cells in diverse tissues exhibit site-specific maturation dynamics across the life span. *Sci Immunol* 2024;9:eadm3954.
 - 53 Cooke KR, Kobzik L, Martin TR, et al. An experimental model of idiopathic pneumonia syndrome after bone marrow transplantation: I. The roles of minor H antigens and endotoxin. *Blood* 1996;88:3230–9.
 - 54 Vydra J, Cosimo E, Lesný P, et al. A Phase I Trial of Allogeneic $\gamma\delta$ T Lymphocytes From Haploidentical Donors in Patients With Refractory or Relapsed Acute Myeloid Leukemia. *Clin Lymphoma Myeloma Leuk* 2023;23:e232–9.
 - 55 Ren J, Liu X, Fang C, et al. Multiplex Genome Editing to Generate Universal CAR T Cells Resistant to PD1 Inhibition. *Clin Cancer Res* 2017;23:2255–66.
 - 56 Basar R, Daher M, Rezvani K. Next-generation cell therapies: the emerging role of CAR-NK cells. *Blood Adv* 2020;4:5868–76.
 - 57 Zhu Y, Smith DJ, Zhou Y, et al. Development of Hematopoietic Stem Cell-Engineered Invariant Natural Killer T Cell Therapy for Cancer. *Cell Stem Cell* 2019;25:542–57.
 - 58 Li Y-R, Zhou Y, Kim YJ, et al. Development of allogeneic HSC-engineered iNKT cells for off-the-shelf cancer immunotherapy. *Cell Rep Med* 2021;2:100449.
 - 59 Dai Z, Zhu Z, Li Z, et al. Off-the-shelf invariant NKT cells expressing anti-PSCA CAR and IL-15 promote pancreatic cancer regression in mice. *J Clin Invest* 2025;135:e179014.
 - 60 Torikai H, Reik A, Liu P-Q, et al. A foundation for universal T-cell based immunotherapy: T cells engineered to express a CD19-specific chimeric-antigen-receptor and eliminate expression of endogenous TCR. *Blood* 2012;119:5697–705.
 - 61 Benjamin R, Graham C, Yallop D, et al. Genome-edited, donor-derived allogeneic anti-CD19 chimeric antigen receptor T cells in paediatric and adult B-cell acute lymphoblastic leukaemia: results of two phase 1 studies. *Lancet* 2020;396:1885–94.
 - 62 Qasim W, Zhan H, Samarasinghe S, et al. Molecular remission of infant B-ALL after infusion of universal TALEN gene-edited CAR T cells. *Sci Transl Med* 2017;9:eaaj2013.
 - 63 Stadtmauer EA, Fraietta JA, Davis MM, et al. CRISPR-engineered T cells in patients with refractory cancer. *Science* 2020;367:eaba7365.
 - 64 Ma S, Caligiuri MA, Yu J. Harnessing Natural Killer Cells for Lung Cancer Therapy. *Cancer Res* 2023;83:3327–39.
 - 65 Mikulak J, Oriolo F, Bruni E, et al. Nkp46-expressing human gut-resident intraepithelial V δ 1 T cell subpopulation exhibits high antitumor activity against colorectal cancer. *JCI Insight* 2019;4:e125884.
 - 66 Nishimoto KP, Barca T, Azameera A, et al. Allogeneic CD20-targeted $\gamma\delta$ T cells exhibit innate and adaptive antitumor activities in preclinical B-cell lymphoma models. *Clin Transl Immunology* 2022;11:e1373.
 - 67 Lee D, Dunn ZS, Guo W, et al. Unlocking the potential of allogeneic V δ 2 T cells for ovarian cancer therapy through CD16 biomarker selection and CAR/IL-15 engineering. *Nat Commun* 2023;14:6942.
 - 68 Wang Y, Ji N, Zhang Y, et al. B7H3-targeting chimeric antigen receptor modification enhances antitumor effect of V γ 9V δ 2 T cells in glioblastoma. *J Transl Med* 2023;21:672.
 - 69 Fulford TS, Soliman C, Castle RG, et al. V γ 9V δ 2 T cells recognize butyrophilin 2A1 and 3A1 heteromers. *Nat Immunol* 2024;25:1355–66.
 - 70 Uldrich AP, Le Nours J, Pellicci DG, et al. CD1d-lipid antigen recognition by the $\gamma\delta$ TCR. *Nat Immunol* 2013;14:1137–45.
 - 71 Spada FM, Grant EP, Peters PJ, et al. Self-recognition of CD1 by gamma/delta T cells: implications for innate immunity. *J Exp Med* 2000;191:937–48.
 - 72 Luoma AM, Castro CD, Mayassi T, et al. Crystal structure of V δ 1 T cell receptor in complex with CD1d-sulfatide shows MHC-like recognition of a self-lipid by human $\gamma\delta$ T cells. *Immunity* 2013;39:1032–42.
 - 73 Bauer S, Groh V, Wu J, et al. Activation of NK cells and T cells by NKG2D, a receptor for stress-inducible MICA. *Science* 1999;285:727–9.
 - 74 Poggi A, Venturino C, Catellani S, et al. Vdelta1 T lymphocytes from B-CLL patients recognize ULBP3 expressed on leukemic B cells and up-regulated by trans-retinoic acid. *Cancer Res* 2004;64:9172–9.
 - 75 Knight A, Mackinnon S, Lowdell MW. Human Vdelta1 gamma-delta T cells exert potent specific cytotoxicity against primary multiple myeloma cells. *Cytotherapy* 2012;14:1110–8.
 - 76 Wolf FA, Angerer P, Theis FJ. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biol* 2018;19:15.
 - 77 Schroeder MA, Ashami K, Staser K. Modeling Chronic Graft Versus Host Disease in Mice Using Allogeneic Bone Marrow and Splenocyte Transfer. *Curr Protoc Pharmacol* 2018;83:e47.